



Structure and Bridge Division



VDOT Supplement to the AASHTO Manual for Bridge Element Inspection

January 2026

NOTES

PREFACE AND ACKNOWLEDGEMENTS

The previous version of the VDOT Supplement to the AASHTO Manual for Bridge Element Inspection, released in January 2016, has been revised to reflect revisions to the AASHTO Manual for Bridge Element Inspection (AASHTO MBEI) current as of the publication of the AASHTO MBEI, Second Edition (2019) with 2022, 2024, and 2025 Interim Revisions and was prepared as a collaborative effort between the following organizations and acknowledges the contributions of each.



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INTRODUCTION AND BACKGROUND

This manual will serve as the Virginia Supplement to the AASHTO Manual for Bridge Element Inspection with revisions to this manual current as of the publication of the AASHTO Manual for Bridge Element Inspection (MBEI), Second Edition (2019) with 2022, 2024, and 2025 Interim Revisions. This manual includes information concerning VDOT Agency Defined Elements, VDOT Defects, and VDOT guidance.

A bridge management system (BMS) provides the tools to consider where and when to best spend funds. It provides an initial indication of the best action to take for each bridge to promote the health of the network.

Using a bridge management system, a transportation agency should be able to:

- (1) predict deterioration with and without intervening actions;
- (2) identify feasible actions to improve condition, safety, or the ability of an element to function as intended;
- (3) estimate costs and savings;
- (4) determine maintenance strategies;
- (5) optimize a program over a specified period with limited funds; and
- (6) generate reports quantifying bridge needs for use by legislative budget makers.

Between 1995 and prior to April 2015, VDOT had been collecting element level inspection data using the AASHTO Commonly Recognized Elements (CoRe elements), VDOT Defined Elements and Smart Flags. This collection of Element Level Data was not required by the National Bridge Inspection Standards (NBIS).

Title 23 of the United States Code, as amended by the MAP-21 legislation in 2012, required each state to report Element Level Data for all bridges on the National Highway System (NHS). See Appendix A.

The FHWA issued the Specification for the National Bridge Inventory Bridge Elements on December 16, 2013. This Specification provided the framework to support the collection and reporting of element level bridge condition data to the FHWA, contained a listing of the elements for which data has been collected by the FHWA and made reference to the AASHTO MBEI, First Edition, for element descriptions, quantity calculations, and condition states. As of January 1, 2026, National Bridge Inventory (NBI) data and reporting has transitioned to a new data standard, guidance, procedures, and web application using The Specification for the National Bridge Inventory (SNBI) to satisfy the requirements of FHWA's Implementation Memo "Implementation of the Specifications for the National Bridge Inventory" dated May 25, 2022. A link to the SNBI provided as follows:
<https://www.fhwa.dot.gov/bridge/nbi.cfm>

The AASHTO elements are the basis for the system for categorizing bridge conditions as outlined in the AASHTO MBEI.

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The AASHTO MBEI includes three Element types (shown below) and Defects associated with each Element:

- National Bridge Elements (NBEs) - which are Elements that are reported to FHWA.
- Bridge Management Elements (BMEs) which are Elements used by the individual states some of which are and some are not reported to FHWA. See Pages 3 and 4 for a list of elements that are reported to FHWA.
- Agency Defined Elements (ADEs) which are state elements used by the individual states but are not reported to FHWA.

Twenty (20) VDOT Agency Defined Elements and ten (10) VDOT Defects have been defined. This document will provide guidance for the coding of the VDOT elements and the VDOT defects.

The 20 VDOT Agency Defined Elements are sub-divided into the following.

- 1 NBE Sub-Elements
- 0 BME Sub-Elements
- 19 ADE Elements

In addition to the condition of a bridge element, its environment and the effects of traffic may govern its rate of deterioration. VDOT has decided to place all elements in the “low” environment (Environment 2), which will not adversely affect the condition of the element.

NBE/BME ELEMENTS TO BE REPORTED TO FHWA

Element	Units	Element Number					
		Steel	Prestressed Concrete	Reinforced Concrete	Timber	Masonry	Other
Deck/Slab							
Deck	SF		13	12	31		60
Open Grid Deck	SF	28					
Concrete Filled Grid Deck	SF	29					
Corrugated or Orthotropic Deck	SF	30					
Slab	SF		39	38	54		65
Top Flange	SF		15	16			
Superstructure							
Closed Web/Box Girder	LF	102	104	105			106
Girder/Beam	LF	107	109	110	111		112
Stringer	LF	113	115	116	117		118
Truss	LF	120			135		136
Arch	LF	141	143	144	146	145	142
Main Cable	LF	147					
Secondary Cable	EA	148					149
Floor Beam	LF	152	154	155	156		157
Pin, Pin and Hanger Assembly	EA	161					
Gusset Plate	EA	162					
Primary Cable Saddles	EA	164					
Cable Anchorage Socket or Assembly	EA	165					
Substructure							
Column	EA	202	204	205	206		203
Column Tower (Trestle)	LF	207		237	208	238	
Pier Wall	LF			210	212	213	211
Abutment	LF	219		215	216	217	218
Pile Cap/Footing	LF			220			
Pile	EA	225	226	227	228		229
Pier Cap	LF	231	233	234	235		236
Anchorage Chamber Walls	LF	222		221		223	
Reinforced Concrete Anchorage Block	EA			224			
Culvert							
Culvert	LF	240	245	241	242	244	243
Bridge Rail							
Bridge Rail	LF	330*		331	332	334	333

*Element 330-Metal Bridge Rail may include steel/aluminum rails.

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NBE/BME ELEMENTS TO BE REPORTED TO FHWA

Element	Units	Element Number
Joint		
Strip Seal	LF	300
Pourable	LF	301
Compression	LF	302
Assembly with Seal	LF	303
Open	LF	304
Assembly without Seal	LF	305
Other	LF	306
Bearing		
Elastomeric	EA	310
Movable (roller, sliding, etc.)	EA	311
Enclosed/Concealed	EA	312
Fixed	EA	313
Pot	EA	314
Disk	EA	315
Other	EA	316
Wearing Surfaces and Protective Coatings		
Wearing Surfaces	SF	510
Steel Protective Coating	SF	515
Concrete Protective Coating	SF	521

VDOT AGENCY DEFINED ELEMENTS (ADEs)

VDOT Agency Defined Elements are denoted by 800 series element numbers.

The numbering system for the VDOT Agency Defined Elements is shown below:

- 801 – 810 – Deck Elements
- 811 – 820 – Superstructure Elements
- 821 – 830 – Substructure Elements
- 831 – 840 – Culvert and Structure Under Fill Elements
- 841 – 850 – Joint Elements
- 851 – 860 – Slope and Channel Elements
- 861 – 880 – Reserved for Future Use
- 881 – 889 – Protective Elements

In the following charts much information is contained including many parent/child associations.

VDOT Agency Defined Elements	Units	ADE Number	Sub – Element (Child) to NBE	NBE Number	Sub – Element (Child) to BME	BME Number	Comments
Sidewalk/Median	SF	801	No	N/A	No	N/A	
Deck Drains	EA	802	No	N/A	No	N/A	
Beam/Girder End	EA	811	No	N/A	No	N/A	Only Elements 886 and 887 may be used in conjunction with this element.
Steel Diaphragms/Cross Frames for Curved Girder Systems	LF	813	Yes	107	No	N/A	Code for all curved steel girders. Element 515 may be used with this element.
Metal Wingwall	EA	822	No	N/A	No	N/A	
Reinforced Concrete Wingwall	EA	824	No	N/A	No	N/A	
Timber Wingwall	EA	826	No	N/A	No	N/A	
Masonry Wingwall	EA	828	No	N/A	No	N/A	
MSE Wall (50' Maximum per Wing)	LF	830	No	N/A	No	N/A	Wingwalls >50 ft. will be roadway items.
Culvert/Structure Under Fill Endwall/Headwall	EA	831	No	N/A	No	N/A	
Roadway Over Fill	EA	833	No	N/A	No	N/A	Code for condition of any roadway material over any type of fill.
Link Slab	EA	843	No	N/A	No	N/A	Joint closures at piers.
Slab Extension	EA	844	No	N/A	No	N/A	Joint closures at abutments.
Joint Effectiveness	EA	845	No	N/A	No	N/A	Code for all joints.
Unprotected Slope	EA	851	No	N/A	No	N/A	
Protected Slope Paved	EA	852	No	N/A	No	N/A	
Protected Slope Riprap	EA	853	No	N/A	No	N/A	
Channel	EA	854	No	N/A	No	N/A	
Beam/Girder End Protective Coating – Steel	EA	886	No	N/A	No	N/A	Element may only be used in combination with Element 811.
Beam/Girder End Protective Coating – Concrete	EA	887	No	N/A	No	N/A	Element may only be used in combination with Element 811.

NOTE: See 'VDOT ADEs Condition State Definitions – with Associated Defects' for additional coding instructions

DESCRIPTIONS OF VDOT AGENCY DEFINED ELEMENTS

ELEMENT NUMBER	TITLE	DESCRIPTION
801	Sidewalk/Median	This element defines a raised/elevated sidewalk/median on a deck element that is 3 feet or more in width. See Page 68 for additional guidance. Use appropriate material defects.
802	Deck Drains	This element defines deck drains (scuppers), grates, and downspouts and does not apply to drainage off the end of a bridge.
811	Beam/Girder End	This element defines the last 5 feet of beam/girder ends under a joint only (Elements 102, 104, 105, 107, 109, 110) and should not be coded at integral abutments, continuous span/deck locations, joint elimination retrofits, or similar. Measurement shall begin at the end of the beam/girder and continue toward the center of the span. Only Elements 886 and 887 may be used in conjunction with this element. Use appropriate material defects.
813	Steel Diaphragms/ Cross Frames for Curved Girder Systems	This element defines steel diaphragms/cross frames for curved girder systems regardless of protective systems. Element 515 may be used with this element. Use appropriate material defects.
822	Metal Wingwall	This element defines metal wingwalls and includes the sheet material retaining the embankment. Use steel defects regardless of metal type. Corrosion includes oxidation. This is for all wingwalls regardless of protective systems.
824	Reinforced Concrete Wingwall	This element defines reinforced concrete wingwalls. This includes the material retaining the embankment. This is for all reinforced concrete wingwalls regardless of protective systems. Use reinforced concrete defects.
826	Timber Wingwall	This element defines timber wingwalls. This includes the sheet material retaining the embankment. This is for all wingwalls regardless of protective system. Use timber defects.
828	Masonry Wingwall	This element defines those wingwalls constructed of block or stone and includes gabion baskets. The block or stone may be placed with or without mortar. This is for all wingwalls regardless of protective systems. Use masonry defects.
830	MSE Wall	This element defines those walls constructed with facing and various combinations of geotechnical materials and reinforcement materials. Total quantity shall be taken as the portion of the MSE wall underneath the structure from bridge width out-to-out and to also include up to 50' of additional length of each "wing" portion regardless of the controlling break point as shown in the Substructure Element guidance on Page 60.
831	Culvert/Structure Under Fill Endwall/ Headwall	This element defines culvert/structure under fill endwalls/headwalls of any material. Use appropriate material defects.

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DESCRIPTIONS OF VDOT AGENCY DEFINED ELEMENTS

ELEMENT NUMBER	TITLE	DESCRIPTION
833	Roadway Over Fill	This element defines the condition of any roadway material over any type of fill.
843	Link Slab	This element defines joint closures constructed at piers. Defect quantities should also be included in the overall deck quantities as this element is part of the deck system.
844	Slab Extension	This element defines joint closures constructed at abutments. Defect quantities should also be included in the overall deck quantities as this element is part of the deck system.
845	Joint Effectiveness	This element defines the effectiveness of expansion joints. There are two condition states for this element.
851	Unprotected Slope	This element defines slopes under bridges that are NOT protected by an erosion resistant material.
852	Protected Slope – Paved	This element defines slopes under a bridge that are protected by an erosion resistant material such as precast concrete blocks or cast-in-place concrete.
853	Protected Slope – Riprap	This element defines slopes under a bridge that are protected by an erosion resistant stone material such as riprap. This element includes combination concrete slab and stone riprap.
854	Channel	This element defines the channel, stream control devices, and channel protective devices such as channel armoring, gabions, grout filled bags, and other protective devices that are not on the slope under the bridge as well as to indicate any drift that may be present.
886	Beam/Girder End Protective Coating – Steel	This element is for the last 5 feet of steel beam/girder ends under a joint only (Elements 102 and 107) that have a protective coating such as paint, galvanization, weathering steel patina or other top coat steel corrosion inhibitor and should not be coded at integral abutments, continuous span/deck locations, joint elimination retrofits, or similar. Measurement shall begin at the end of the beam/girder and continue toward the center of the span. Element may only be used in combination with Element 811.
887	Beam/Girder End Protective Coating – Concrete	This element is for the last 5 feet of concrete beam/girder ends under a joint only (Elements 104, 105, 109, and 110) that have a protective coating such as epoxy or any other top coat barrier that protects concrete from deterioration and reinforcing steel corrosion and should not be coded at integral abutments, continuous span/deck locations, joint elimination retrofits, or similar. Measurement shall begin at the end of the beam/girder and continue toward the center of the span. Element may only be used in combination with Element 811.

VDOT AGENCY DEFINED DEFECTS

VDOT DEFECT NUMBER	TITLE	DESCRIPTION	VDOT ASSOCIATED ELEMENT
2000	Deck Drainage System Effectiveness	Indicates the effectiveness of deck drainage system.	802
2399	Joint Effectiveness	Indicates the number of joints in functional and non-functional condition.	845
6500	Debris	Debris restricting channel flow.	854
6501	Facing Panel Displacement/ Separation	Displacement or separation of facing panels for MSE walls	830
6502	Facing Panel Bulging	Bulging of facing panels for MSE walls.	830
6503	Facing Panel Joint Opening	Joint openings for MSE walls	830
6505	Unprotected Slope Effectiveness	Indicates the effectiveness of unprotected slopes.	851
6506	Paved Slope Effectiveness	Indicates effectiveness of paved slopes	852
6507	Loss of Riprap	Indicates the loss of riprap stone on slopes.	853
6508	Effectiveness of Stream Control and Channel Protective Devices	Indicates the effectiveness of stream control and channel protective devices.	854

DEFECTS FOR REINFORCED CONCRETE (FROM MBEI)

DEFECTS		CS 1 GOOD	CS 2 FAIR	CS 3 POOR	CS 4 SEVERE
1080	Delamination/Spall/ Patched Area	None.	Delaminated. Spall 1 in. or less deep or 6 in. or less in diameter. Patched area that is sound.	Spall greater than 1 in. deep or greater than 6 in. diameter. Patched area that is unsound or showing distress. Does not warrant structural review.	The condition warrants a structural review to determine the effect on strength or serviceability of the element or bridge; OR a structural review has been completed and the defects impact strength or serviceability of the element or bridge.
1090	Exposed Rebar	None.	Present without measurable section loss.	Present with measurable section loss but does not warrant structural review.	
1120	Efflorescence/Rust Staining	None.	Efflorescence without rust staining.	Efflorescence with rust staining.	
1130	Cracking in Reinforced Concrete	Insignificant cracks or moderate-width cracks that have been sealed.	Unsealed moderate-width cracks, or unsealed moderate pattern (map) cracking.	Wide cracks or heavy pattern (map) cracking.	
1190	Abrasion/Wear	No abrasion or wearing.	Abrasion or wearing has exposed coarse aggregate but the aggregate remains secure in the concrete.	Coarse aggregate is loose or has popped out of the concrete matrix due to abrasion or wear.	
1900	Distortion	None.	Distortion that does not require mitigation or mitigated distortion.	Distortion that requires mitigation that has not been addressed but does not warrant structural review.	
4000	Settlement	None.	Exists within tolerable limits or arrested with no observed structural distress.	Exceeds tolerable limits but does not warrant structural review.	
6000	Scour	None.	Exists within tolerable limits or has been arrested with effective countermeasures.	Exceeds tolerable limits but is less than the critical limits determined by scour evaluation and does not warrant structural review.	

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DEFECTS FOR REINFORCED CONCRETE (FROM MBEI)

DEFECTS		CS 1	CS 2	CS 3	CS 4
		GOOD	FAIR	POOR	SEVERE
7000	Damage	Not applicable.	The element has impact damage. The specific damage caused by the impact has been captured in CS 2 under the appropriate material defect entry.	The element has impact damage. The specific damage caused by the impact has been captured in CS 3 under the appropriate material defect entry.	The element has impact damage. The specific damage caused by the impact has been captured in CS 4 under the appropriate material defect entry.

DEFECTS FOR PRESTRESSED CONCRETE (FROM MBEI)

DEFECTS		CS 1 GOOD	CS 2 FAIR	CS 3 POOR	CS 4 SEVERE
1080	Delamination/Spall/ Patched Area	None.	Delaminated. Spall 1 in. or less deep or 6 in. or less in diameter. Patched area that is sound.	Spall greater than 1 in. deep or greater than 6 in. diameter. Patched area that is unsound or showing distress. Does not warrant structural review.	The condition warrants a structural review to determine the effect on strength or serviceability of the element or bridge; OR a structural review has been completed and the defects impact strength or serviceability of the element or bridge.
1090	Exposed Rebar	None.	Present without measurable section loss.	Present with measurable section loss but does not warrant structural review.	
1100	Exposed Prestressing	None.	Present without section loss.	Present with section loss but does not warrant structural review.	
1120	Efflorescence/Rust Staining	None.	Efflorescence without rust staining.	Efflorescence with rust staining.	
1110	Cracking in Prestressed Concrete	Insignificant cracks or moderate-width cracks that have been sealed.	Unsealed moderate-width cracks, or unsealed moderate pattern (map) cracking.	Wide cracks or heavy pattern (map) cracking.	
1190	Abrasion/Wear	No abrasion or wearing.	Abrasion or wearing has exposed coarse aggregate but the aggregate remains secure in the concrete.	Coarse aggregate is loose or has popped out of the concrete matrix due to abrasion or wear.	
1900	Distortion	None.	Distortion that does not require mitigation or mitigated distortion.	Distortion that requires mitigation that has not been addressed but does not warrant structural review.	
4000	Settlement	None.	Exists within tolerable limits or arrested with no observed structural distress.	Exceeds tolerable limits but does not warrant structural review.	

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DEFECTS FOR PRESTRESSED CONCRETE (FROM MBEI)

DEFECTS		CS 1	CS 2	CS 3	CS 4
		GOOD	FAIR	POOR	SEVERE
6000	Scour	None.	Exists within tolerable limits or has been arrested with effective countermeasures.	Exceeds tolerable limits but is less than the critical limits determined by scour evaluation and does not warrant structural review.	
7000	Damage	Not applicable.	The element has impact damage. The specific damage caused by the impact has been captured in CS 2 under the appropriate material defect entry.	The element has impact damage. The specific damage caused by the impact has been captured in CS 3 under the appropriate material defect entry.	The element has impact damage. The specific damage caused by the impact has been captured in CS 4 under the appropriate material defect entry.

DEFECTS FOR STEEL (FROM MBEI)

DEFECTS		CS 1 GOOD	CS 2 FAIR	CS 3 POOR	CS 4 SEVERE
1000	Corrosion	None.	Freckled rust. Corrosion of the steel has initiated.	Section loss is evident, or pack rust is present but does not warrant structural review.	The condition warrants a structural review to determine the effect on strength or serviceability of the element or bridge; OR a structural review has been completed and the defects impact strength or serviceability of the element or bridge.
1010	Cracking in Steel	None.	Crack that has self-arrested or has been arrested with effective arrest holes, doubling plates, or similar.	Identified crack that is not arrested but does not warrant structural review.	
1020	Connection	Connection is in place and functioning as intended.	Loose fasteners or pack rust without distortion is present but the connection is in place and functioning as intended.	Missing bolts, rivets, or fasteners; broken welds; or pack rust with distortion but does not warrant structural review.	
1030	Misalignment	None.	Tolerable misalignment.	Significant misalignment but does not warrant structural review.	
1200	Wear	None.	Minor wear or fretting corrosion.	Moderate to significant wear or fretting corrosion is present but does not warrant structural review.	
1900	Distortion	None.	Distortion that does not require mitigation or mitigated distortion.	Distortion that requires mitigation that has not been addressed but does not warrant structural review.	
4000	Settlement	None.	Exists within tolerable limits or arrested with no observed structural distress.	Exceeds tolerable limits but does not warrant structural review.	
6000	Scour	None.	Exists within tolerable limits or has been arrested with effective countermeasures.	Exceeds tolerable limits but is less than the critical limits determined by scour evaluation and does not warrant structural review.	

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DEFECTS FOR STEEL (FROM MBEI)

DEFECTS		CS 1	CS 2	CS 3	CS 4
		GOOD	FAIR	POOR	SEVERE
7000	Damage	Not applicable.	The element has impact damage. The specific damage caused by the impact has been captured in CS 2 under the appropriate material defect entry.	The element has impact damage. The specific damage caused by the impact has been captured in CS 3 under the appropriate material defect entry.	The element has impact damage. The specific damage caused by the impact has been captured in CS 4 under the appropriate material defect entry.

DEFECTS FOR TIMBER (FROM MBEI)

DEFECTS		CS 1 GOOD	CS 2 FAIR	CS 3 POOR	CS 4 SEVERE
1020	Connection	Connection is in place and functioning as intended.	Loose fasteners or pack rust without distortion is present but the connection is in place and functioning as intended.	Missing bolts, rivets, or fasteners; broken welds; or pack rust with distortion but does not warrant structural review.	The condition warrants a structural review to determine the effect on strength or serviceability of the element or bridge; OR a structural review has been completed and the defects impact strength or serviceability of the element or bridge.
1140	Decay/Section Loss	None.	Affects less than 10% of the member section.	Affects 10% or more of the member but does not warrant structural review.	
1150	Check/Shake	Surface penetration less than 5% of the member thickness regardless of location.	Penetrates 5% to 50% of the thickness of the member and not in a tension zone.	Penetrates more than 50% of the thickness of the member or penetrates more than 5% of the thickness of the member in the tension zone. Does not warrant structural review.	
1160	Cracking in Timber	None.	Crack has been arrested through effective measures.	Identified crack that is not arrested but does not require structural review.	
1170	Split/Delamination in Timber	None.	Length less than the member depth or arrested with effective actions taken to mitigate.	Length equal to or greater than the member depth but does not require structural review.	
1180	Abrasion/Wear in Timber	None or no measurable section loss.	Section loss less than 10% of the member thickness.	Section loss 10% or more of the member thickness but does not warrant structural review.	
4000	Settlement	None.	Exists within tolerable limits or arrested with no observed structural distress.	Exceeds tolerable limits but does not warrant structural review.	

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DEFECTS FOR TIMBER (FROM MBEI)

DEFECTS		CS 1	CS 2	CS 3	CS 4
		GOOD	FAIR	POOR	SEVERE
6000	Scour	None.	Exists within tolerable limits or has been arrested with effective countermeasures.	Exceeds tolerable limits but is less than the critical limits determined by scour evaluation and does not warrant structural review.	
7000	Damage	Not applicable.	The element has impact damage. The specific damage caused by the impact has been captured in CS 2 under the appropriate material defect entry.	The element has impact damage. The specific damage caused by the impact has been captured in CS 3 under the appropriate material defect entry.	The element has impact damage. The specific damage caused by the impact has been captured in CS 4 under the appropriate material defect entry.

DEFECTS FOR MASONRY (FROM MBEI)

DEFECTS		CS 1 GOOD	CS 2 FAIR	CS 3 POOR	CS 4 SEVERE
1080	Delamination/Spall/ Patched Area	None.	Delaminated. Spall 1 in. or less deep or 6 in. or less in diameter. Patched area that is sound.	Spall greater than 1 in. deep or greater than 6 in. diameter. Patched area that is unsound or showing distress. Does not warrant structural review.	The condition warrants a structural review to determine the effect on strength or serviceability of the element or bridge; OR a structural review has been completed and the defects impact strength or serviceability of the element or bridge.
1120	Efflorescence/Rust Staining	None.	Efflorescence without rust staining.	Efflorescence with rust staining.	
1610	Mortar Breakdown in Masonry	None.	Cracking or voids in less than 10% of joints.	Cracking or voids in 10% or more of the joints.	
1620	Split/Spall in Masonry	None	Block or stone has split or spalled with no shifting.	Block or stone has split or spalled with shifting but does not warrant a structural review.	
1630	Patched Area in Masonry	None.	Sound patch.	Unsound patch.	
1640	Masonry Displacement	None.	Block or stone has shifted slightly out of alignment.	Block or stone has shifted significantly out of alignment or is missing but does not warrant structural review.	
1900	Distortion	None.	Distortion that does not require mitigation or mitigated distortion.	Distortion that requires mitigation that has not been addressed but does not warrant structural review.	
4000	Settlement	None.	Exists within tolerable limits or arrested with no observed structural distress.	Exceeds tolerable limits but does not warrant structural review.	
6000	Scour	None.	Exists within tolerable limits or has been arrested with effective countermeasures.	Exceeds tolerable limits but is less than the critical limits determined by scour evaluation and does not warrant structural review.	

(continues)

(continued)

DEFECTS FOR MASONRY (FROM MBEI)

DEFECTS		CS 1	CS 2	CS 3	CS 4
		GOOD	FAIR	POOR	SEVERE
7000	Damage	Not applicable.	The element has impact damage. The specific damage caused by the impact has been captured in CS 2 under the appropriate material defect entry.	The element has impact damage. The specific damage caused by the impact has been captured in CS 3 under the appropriate material defect entry.	The element has impact damage. The specific damage caused by the impact has been captured in CS 4 under the appropriate material defect entry.

DEFECTS FOR OTHER MATERIAL ELEMENTS (FROM MBEI)

DEFECTS		CS 1 GOOD	CS 2 FAIR	CS 3 POOR	CS 4 SEVERE
1000	Corrosion	None.	Freckled rust. Corrosion of the steel has initiated.	Section loss is evident, or pack rust is present but does not warrant structural review.	The condition warrants a structural review to determine the effect on strength or serviceability of the element or bridge; OR a structural review has been completed and the defects impact strength or serviceability of the element or bridge.
1010	Cracking	None.	Crack that has self-arrested or has been arrested with effective arrest holes, doubling plates, or similar.	Identified crack that is not arrested but does not warrant structural review.	
1020	Connection	Connection is in place and functioning as intended.	Loose fasteners or pack rust without distortion is present but the connection is in place and functioning as intended.	Missing bolts, rivets, or fasteners; broken welds; or pack rust with distortion but does not warrant structural review.	
1080	Delamination/Spall/ Patched Area	None.	Delaminated. Spall 1 in. or less deep or 6 in. or less in diameter. Patched area that is sound.	Spall greater than 1 in. deep or greater than 6 in. diameter. Patched area that is unsound or showing distress. Does not warrant structural review.	
1120	Efflorescence/Rust Staining	None.	Efflorescence without rust staining.	Efflorescence with rust staining.	
1130	Cracking in Other Materials	Insignificant cracks or moderate-width cracks that have been sealed.	Unsealed moderate-width cracks, or unsealed moderate pattern (map) cracking.	Wide cracks or heavy pattern (map) cracking.	
1220	Deterioration	None.	Initiated breakdown or deterioration.	Significant deterioration or breakdown but does not warrant structural review.	
1900	Distortion	None.	Distortion that does not require mitigation or mitigated distortion.	Distortion that requires mitigation that has not been addressed but does not warrant structural review.	

(continues)

(continued)

DEFECTS FOR OTHER MATERIAL ELEMENTS (FROM MBEI)

DEFECTS		CS 1	CS 2	CS 3	CS 4
		GOOD	FAIR	POOR	SEVERE
4000	Settlement	None.	Exists within tolerable limits or arrested with no observed structural distress.	Exceeds tolerable limits but does not warrant structural review.	
6000	Scour	None.	Exists within tolerable limits or has been arrested with effective countermeasures.	Exceeds tolerable limits but is less than the critical limits determined by scour evaluation and does not warrant structural review.	
7000	Damage	Not applicable.	The element has impact damage. The specific damage caused by the impact has been captured in CS 2 under the appropriate material defect entry.	The element has impact damage. The specific damage caused by the impact has been captured in CS 3 under the appropriate material defect entry.	The element has impact damage. The specific damage caused by the impact has been captured in CS 4 under the appropriate material defect entry.

VDOT ADEs CONDITION STATE DEFINITIONS – with Associated Defects

VDOT ELEMENT NUMBER	802	VDOT ELEMENT NAME	DECK DRAINS
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DESCRIPTION	This element defines deck drains/scuppers, grates, and downspouts and does not apply to drainage off the end of a bridge. The quantity will always be one, even if there are dissimilar drainage types.
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UNIT OF MEASURE	EA
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ALL DECK DRAINS					
DEFECT		CS 1 GOOD	CS 2 FAIR	CS 3 POOR	CS 4 SEVERE
2000	Deck Drainage System Effectiveness	Grating over scuppers/drains is intact and is functioning as intended. Bridge drainage and/or downspout is adequately terminated. No signs of ponding on the deck as a result of drainage problems.	Scuppers and/or downspouts are clogged; however, there are no signs of ponding on the deck. Downspout is inadequately terminated.	Scuppers and/or downspouts are clogged and there are signs of ponding on the deck, but does not extend into the normal traffic lane.	Broken/missing grates/drainage assembly may pose a hazard to vehicular or pedestrian traffic. Scuppers and/or downspouts are clogged and there are signs of ponding on the deck that extends into the normal traffic lane.
7000	Damage	Not applicable.	The element has impact damage. The specific damage caused by the impact has been captured in CS 2 under the appropriate material defect entry.	The element has impact damage. The specific damage caused by the impact has been captured in CS 3 under the appropriate material defect entry.	The element has impact damage. The specific damage caused by the impact has been captured in CS 4 under the appropriate material defect entry.

VDOT ELEMENT NUMBER	830
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VDOT ELEMENT NAME	MSE WALL
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DESCRIPTION	This element defines those walls constructed with facing and various combinations of geotechnical materials and reinforcement materials. Total quantity shall be taken as the portion of the MSE wall underneath the structure from bridge width out-to-out and to also include up to 50' of additional length of each "wing" portion regardless of the controlling break point as shown in the Substructure Element guidance on Page 60.
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UNIT OF MEASURE	LF
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MSE WALL					
DEFECT		CS 1 GOOD	CS 2 FAIR	CS 3 POOR	CS 4 SEVERE
1080	Delamination/Spall/ Patched Area	None.	Delaminated. Spall 1 in. or less deep or 6 in. or less in diameter. Patched area that is sound.	Spall greater than 1 in. deep or greater than 6 in. diameter. Patched area that is unsound or showing distress. Does not warrant structural review.	The condition warrants a structural review to determine the effect on strength or serviceability of the element or bridge; OR a structural review has been completed and the defects impact strength or serviceability of the element or bridge.
1090	Exposed Rebar	None.	Present without measurable section loss.	Present with measurable section loss but does not warrant structural review.	
1120	Efflorescence/Rust Staining	None.	Efflorescence without rust staining.	Efflorescence with rust staining.	
1130	Cracking in Reinforced Concrete	Insignificant cracks or moderate-width cracks that have been sealed.	Unsealed moderate-width cracks, or unsealed moderate pattern (map) cracking.	Wide cracks or heavy pattern (map) cracking.	
4000	Settlement	None.	Exists within tolerable limits or arrested with no observed structural distress.	Exceeds tolerable limits but does not warrant structural review.	
6000	Scour	None.	Exists within tolerable limits or has been arrested with effective countermeasures.	Exceeds tolerable limits but is less than the critical limits determined by scour evaluation and does not warrant structural review.	

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VDOT ELEMENT NUMBER	830
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VDOT ELEMENT NAME	MSE WALL
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MSE WALL					
DEFECT		CS 1	CS 2	CS 3	CS 4
		GOOD	FAIR	POOR	SEVERE
6501	Facing Panel Displacement/Separation	None.	Facing panels have shifted slightly out of alignment.	Facing panels have shifted significantly out of alignment or are missing and do not warrant structural review.	
6502	Facing Panel Bulging	None.	Facing panels have bulged slightly.	Facing panels have bulged significantly and do not warrant structural review.	
6503	Facing Panel Joint Opening	None.	Facing panel joints have minor openings.	Facing panel joints have significant joint openings and do not warrant structural review.	
7000	Damage	Not applicable.	The element has impact damage. The specific damage caused by the impact has been captured in CS 2 under the appropriate material defect entry.	The element has impact damage. The specific damage caused by the impact has been captured in CS 3 under the appropriate material defect entry.	The element has impact damage. The specific damage caused by the impact has been captured in CS 4 under the appropriate material defect entry.

VDOT ELEMENT NUMBER	833	VDOT ELEMENT NAME	ROADWAY OVER FILL
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DESCRIPTION	This element defines the condition of any roadway material over any type of fill.
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UNIT OF MEASURE	EA
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ROADWAY OVER FILL					
DEFECT		CS 1	CS 2	CS 3	CS 4
		GOOD	FAIR	POOR	SEVERE
3210	Delamination/Spalls/ Patched Area/Pothole	None.	Delaminated. Spall less than 1 in. deep or less than 6 in. in diameter. Patched area that is sound. Partial depth pothole.	Spall 1 in. deep or greater or 6 in. diameter or greater. Patched area that is unsound or showing distress. Full depth pothole.	The condition warrants a structural review to determine the effect on strength or serviceability of the element or bridge; OR a structural review has been completed and the defects impact strength or serviceability of the element or bridge.
3220	Crack	Insignificant cracks or moderate-width cracks that have been sealed.	Unsealed moderate-width cracks, or unsealed moderate pattern (map) cracking.	Wide cracks or heavy pattern (map) cracking.	
3230	Effectiveness	Fully effective.	Substantially effective. Deterioration of the wearing surface has slowed.	Limited effectiveness. Deterioration of the wearing surface has progressed.	
4000	Settlement	None.	Exists within tolerable limits or arrested with no observed structural distress.	Exceeds tolerable limits but does not warrant structural review.	Roadway over fill has excessive settlement and a substantial reduction in speed is required to safely negotiate this section of roadway over the structure.
6000	Scour	None.	Exists within tolerable limits or has been arrested with effective countermeasures.	Exceeds tolerable limits but is less than the critical limits determined by scour evaluation and does not warrant structural review.	Roadway over fill has voids and areas of erosion that may pose a traffic hazard.

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VDOT ELEMENT NUMBER	833	VDOT ELEMENT NAME	ROADWAY OVER FILL
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ROADWAY OVER FILL					
DEFECT		CS 1	CS 2	CS 3	CS 4
		GOOD	FAIR	POOR	SEVERE
7000	Damage	Not applicable.	The element has impact damage. The specific damage caused by the impact has been captured in CS 2 under the appropriate material defect entry.	The element has impact damage. The specific damage caused by the impact has been captured in CS 3 under the appropriate material defect entry.	The element has impact damage. The specific damage caused by the impact has been captured in CS 4 under the appropriate material defect entry.

VDOT ELEMENT NUMBER	843	VDOT ELEMENT NAME	LINK SLAB
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DESCRIPTION	This element defines joint closures constructed at piers. Defect quantities should also be included in the overall deck quantities as this element is part of the deck system.
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UNIT OF MEASURE	EA
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LINK SLAB					
DEFECT		CS 1 GOOD	CS 2 FAIR	CS 3 POOR	CS 4 SEVERE
1080	Delamination/Spall/ Patched Area	None.	Delaminated. Spall 1 in. or less deep or 6 in. or less in diameter. Patched area that is sound.	Spall greater than 1 in. deep or greater than 6 in. diameter. Patched area that is unsound or showing distress. Does not warrant structural review.	The condition warrants a structural review to determine the effect on strength or serviceability of the element or bridge; OR a structural review has been completed and the defects impact strength or serviceability of the element or bridge.
1090	Exposed Rebar	None.	Present without measurable section loss.	Present with measurable section loss but does not warrant structural review.	
1120	Efflorescence/Rust Staining	None.	Efflorescence without rust staining.	Efflorescence with rust staining.	
1130	Cracking in Reinforced Concrete	Insignificant cracks or moderate-width cracks that have been sealed.	Unsealed moderate-width cracks, or unsealed moderate pattern (map) cracking.	Wide cracks or heavy pattern (map) cracking.	
1190	Abrasion/Wear	No abrasion or wearing.	Abrasion or wearing has exposed coarse aggregate but the aggregate remains secure in the concrete.	Coarse aggregate is loose or has popped out of the concrete matrix due to abrasion or wear.	
2310	Leakage	None.	Minimal. Minor dripping through the joint.	Moderate. More than a drip and less than free flow of water.	Free flow of water through joint.

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VDOT ELEMENT NUMBER	843	VDOT ELEMENT NAME	LINK SLAB
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LINK SLAB					
DEFECT		CS 1	CS 2	CS 3	CS 4
		GOOD	FAIR	POOR	SEVERE
7000	Damage	Not applicable.	The element has impact damage. The specific damage caused by the impact has been captured in CS 2 under the appropriate material defect entry.	The element has impact damage. The specific damage caused by the impact has been captured in CS 3 under the appropriate material defect entry.	The element has impact damage. The specific damage caused by the impact has been captured in CS 4 under the appropriate material defect entry.

VDOT ELEMENT NUMBER	844	VDOT ELEMENT NAME	SLAB EXTENSION
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DESCRIPTION	This element defines joint closures constructed at abutments. Defect quantities should also be included in the overall deck quantities as this element is part of the deck system.
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UNIT OF MEASURE	EA
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SLAB EXTENSION					
DEFECT		CS 1 GOOD	CS 2 FAIR	CS 3 POOR	CS 4 SEVERE
1080	Delamination/Spall/ Patched Area	None.	Delaminated. Spall 1 in. or less deep or 6 in. or less in diameter. Patched area that is sound.	Spall greater than 1 in. deep or greater than 6 in. diameter. Patched area that is unsound or showing distress. Does not warrant structural review.	The condition warrants a structural review to determine the effect on strength or serviceability of the element or bridge; OR a structural review has been completed and the defects impact strength or serviceability of the element or bridge.
1090	Exposed Rebar	None.	Present without measurable section loss.	Present with measurable section loss but does not warrant structural review.	
1120	Efflorescence/Rust Staining	None.	Efflorescence without rust staining.	Efflorescence with rust staining.	
1130	Cracking in Reinforced Concrete	Insignificant cracks or moderate-width cracks that have been sealed.	Unsealed moderate-width cracks, or unsealed moderate pattern (map) cracking.	Wide cracks or heavy pattern (map) cracking.	
1190	Abrasion/Wear	No abrasion or wearing.	Abrasion or wearing has exposed coarse aggregate but the aggregate remains secure in the concrete.	Coarse aggregate is loose or has popped out of the concrete matrix due to abrasion or wear.	
2310	Leakage	None.	Minimal. Minor dripping through the joint.	Moderate. More than a drip and less than free flow of water.	Free flow of water through joint.

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VDOT ELEMENT NUMBER	844	VDOT ELEMENT NAME	SLAB EXTENSION
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SLAB EXTENSION					
DEFECT		CS 1	CS 2	CS 3	CS 4
		GOOD	FAIR	POOR	SEVERE
7000	Damage	Not applicable.	The element has impact damage. The specific damage caused by the impact has been captured in CS 2 under the appropriate material defect entry.	The element has impact damage. The specific damage caused by the impact has been captured in CS 3 under the appropriate material defect entry.	The element has impact damage. The specific damage caused by the impact has been captured in CS 4 under the appropriate material defect entry.

VDOT ELEMENT NUMBER	845	VDOT ELEMENT NAME	JOINT EFFECTIVENESS
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DESCRIPTION	This element defines the effectiveness of expansion joints. There are only two Condition States for this element.
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UNIT OF MEASURE	EA
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JOINT EFFECTIVENESS					
DEFECT		CS 1			CS 4
		GOOD			SEVERE
2399	Joint Effectiveness	Joints that are functional and have no leaks at any point along the length of the joint or show no evidence of being compromised.			Joints that are non-functional and have leaks along the length of the joint or show evidence of being compromised.

VDOT ELEMENT NUMBER	851
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VDOT ELEMENT NAME	UNPROTECTED SLOPE
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DESCRIPTION	This element defines slopes under bridges that are <u>NOT</u> protected by an erosion resistant material.
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UNIT OF MEASURE	EA
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UNPROTECTED SLOPE					
DEFECT		CS 1 GOOD	CS 2 FAIR	CS 3 POOR	CS 4 SEVERE
6505	Unprotected Slope Effectiveness	The element shows little or no erosion.	Minor erosion may have occurred. There is no undermining or exposed piles. There has been no erosion that has exposed the bottom of substructure elements.	Moderate erosion of slope is present but is incidental and does not adversely affect the ability of this element to function as intended. The bottom of the substructure elements may be exposed for insignificant lengths with no undermining or exposed piles.	The slope is no longer functioning as intended. The bottom of the substructure elements are exposed for significant lengths. Piles may be exposed. Substructure elements may be undermined.

VDOT ELEMENT NUMBER	852
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VDOT ELEMENT NAME	PROTECTED SLOPE – PAVED
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DESCRIPTION	This element defines slopes under bridges that <u>ARE</u> protected by an erosion resistant material such as precast concrete blocks or cast-in-place concrete.
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UNIT OF MEASURE	EA
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PROTECTED SLOPE – PAVED					
DEFECT		CS 1	CS 2	CS 3	CS 4
		GOOD	FAIR	POOR	SEVERE
6506	Paved Slope Effectiveness	The slope shows little or no deterioration.	Minor cracks, spalls splitting, settlement or joint deterioration may be present, but there is no evidence of undermining of the paved slope.	Moderate deterioration of slope protection. Minor undermining and/or settlement may be present but erosion due to road run-off is still controlled by slope protection.	Advanced deterioration (misalignment, crushing and/or settlement). Undermining of slope protection renders protection largely ineffective.

VDOT ELEMENT NUMBER	853
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VDOT ELEMENT NAME	PROTECTED SLOPE – RIPRAP
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DESCRIPTION	This element defines slopes under bridges that <u>ARE</u> protected by an erosion resistant stone material such as riprap. This element includes combination concrete slab and stone riprap.
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UNIT OF MEASURE	EA
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PROTECTED SLOPE - RIPRAP					
DEFECT		CS 1 GOOD	CS 2 FAIR	CS 3 POOR	CS 4 SEVERE
6507	Loss of Riprap	None.	Less than 10% of slope area exhibits a loss of stones.	10% to 25% of slope area exhibits a loss of stones.	More than 25% of slope area exhibits a loss of stones.

VDOT ELEMENT NUMBER	854
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VDOT ELEMENT NAME	CHANNEL
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DESCRIPTION	This element defines the channel, stream control devices, and channel protective devices such as channel armoring, gabions, grout filled bags, and other protective devices that are not on the slope under the bridge as well as to indicate any drift that may be present.
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UNIT OF MEASURE	EA
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CHANNEL					
DEFECT		CS 1	CS 2	CS 3	CS 4
		GOOD	FAIR	POOR	SEVERE
1220	Deterioration	None.	Initiated breakdown or deterioration.	Significant deterioration or breakdown but does not warrant structural review.	The condition warrants a structural review to determine the effect on strength or serviceability of the element or bridge; OR a structural review has been completed and the defects impact strength or serviceability of the element or bridge.
4000	Settlement	None.	Exists within tolerable limits or arrested with no observed structural distress.	Exceeds tolerable limits but does not warrant structural review.	
6000	Scour	None.	Exists within tolerable limits or has been arrested with effective countermeasures.	Exceeds tolerable limits but is less than the critical limits determined by scour evaluation and does not warrant structural review.	
6500	Debris	None.	Minor amount of debris in channel. Less than 10% of the opening is blocked. Debris is not detrimental to the structure and no action is required.	Moderate amount of debris in channel. 10% to 25% of the opening is blocked. Debris is not detrimental to the structure and should be removed as soon as practicable to prevent damage/additional accumulation.	Large amount of debris in channel. More than 25% of the opening is blocked. Debris is detrimental to the structure and should be removed right away.
6508	Effectiveness of stream control and channel protective devices	Devices are effective with no loss of material.	Devices are effective and up to 10% of device area exhibits a loss of material.	Devices are moderately effective and 10% to 25% of device area exhibits a loss of material.	Devices are not effective and more than 25% of device area exhibits a loss of material.

VDOT ELEMENT NUMBER	886
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VDOT ELEMENT NAME	BEAM/GIRDER END PROTECTIVE COATING – STEEL
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DESCRIPTION	This element is for the last 5 feet of steel beam / girder ends under a joint only (Elements 102 and 107) that have a protective coating such as paint, galvanization, weathering steel patina or other top coat steel corrosion inhibitor and should not be coded at integral abutments, continuous span/deck locations, joint elimination retrofits, or similar. Measurement shall begin at the end of the beam / girder and continue toward the center of the span. Element may only be used in combination with Element 811.
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UNIT OF MEASURE	EA
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BEAM/GIRDER END PROTECTIVE COATING – STEEL					
DEFECT		CS 1 GOOD	CS 2 FAIR	CS 3 POOR	CS 4 SEVERE
3410	Chalking (Steel Protective Coatings)	None.	Surface dulling.	Loss of pigment.	Not applicable.
3420	Peeling/Bubbling/ Cracking (Steel Protective Coatings)	None.	Finish coats only.	Finish and primer coats.	Exposure of bare metal.
3430	Oxide Film Degradation Color/ Texture Adherence (Steel Protective Coatings)	Yellow-orange or light brown for early development. Chocolate-brown to purple-brown for fully developed. Tightly adhered, capable of withstanding hammering or vigorous wire brushing.	Granular texture.	Small flakes, less than 1/2 in. diameter.	Dark black color. Large flakes, 1/2 in. diameter or greater, or laminar sheets or nodules.
3440	Effectiveness (Steel Protective Coatings)	Fully effective.	Substantially effective.	Limited effectiveness.	Failed; no protection of the underlying metal.
7000	Damage	Not applicable.	The element has impact damage. The specific damage caused by the impact has been captured in CS 2 under the appropriate material defect entry.	The element has impact damage. The specific damage caused by the impact has been captured in CS 3 under the appropriate material defect entry.	The element has impact damage. The specific damage caused by the impact has been captured in CS 4 under the appropriate material defect entry.

VDOT ELEMENT NUMBER	887
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VDOT ELEMENT NAME	BEAM/GIRDER END PROTECTIVE COATING – CONCRETE
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DESCRIPTION	This element is for the last 5 feet of concrete beam / girder ends under a joint only (Elements 104, 105, 109, and 110) that have a protective coating such as epoxy or any other top coat barrier that protects concrete from deterioration and reinforcing steel corrosion and should not be coded at integral abutments, continuous span/deck locations, joint elimination retrofits, or similar. Measurement shall begin at the end of the beam / girder and continue toward the center of the span. Element may only be used in combination with Element 811.
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UNIT OF MEASURE	EA
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BEAM/GIRDER END PROTECTIVE COATING – CONCRETE					
DEFECT		CS 1 GOOD	CS 2 FAIR	CS 3 POOR	CS 4 SEVERE
3510	Wear (Concrete Protective Coatings)	None.	Underlying concrete not exposed; coating showing wear from UV exposure; friction course missing.	Underlying concrete is not exposed; thickness of the coating is reduced.	Underlying concrete exposed. Protective coating no longer effective.
3540	Effectiveness (Concrete Protective Coatings)	Fully effective.	Substantially effective.	Limited effectiveness.	The protective system has failed or is no longer effective.
7000	Damage	Not applicable.	The element has impact damage. The specific damage caused by the impact has been captured in CS 2 under the appropriate material defect entry.	The element has impact damage. The specific damage caused by the impact has been captured in CS 3 under the appropriate material defect entry.	The element has impact damage. The specific damage caused by the impact has been captured in CS 4 under the appropriate material defect entry.

OVERLAPPING DEFECTS

Defects will often overlap. The most obvious case is where two defects occur in the exact same place, a classic example would be exposed rebar in a spall. A less obvious case would be where two defects occur next to each other in the same area. An example would be a one foot section of a girder with cracking and a spall, but the cracking doesn't run through the spall. We can't count this one foot area twice, once for the spall and once for the cracking. That would give us two feet of defects where we have only one foot of element.

In cases where defects overlap, we can only report one defect. The following will assist the inspector in deciding which defect shall be reported. When there are two or more defects in the same area (overlap) then the one in the worst condition state overrides the others. If the worst defect in an area is in Condition State 3, then that portion of the element is in Condition State 3 regardless of how many other Condition State 2 defects share that space with the Condition State 3 defect. In any given area where multiple defects exist only the worst defect will be 'rolled-up' for that area.

This rule works fine until a situation occurs when more than one defect falls into the same condition state in an area i.e. there is a "tie". VDOT has created a hierarchy for these situations to aid the inspector in deciding which defect controls in a tie. This hierarchy should be viewed as a guideline in deciding which defect is the most "important" in any given situation, not as a set of hard and fast rules. Ultimately, inspectors must use their own judgment.

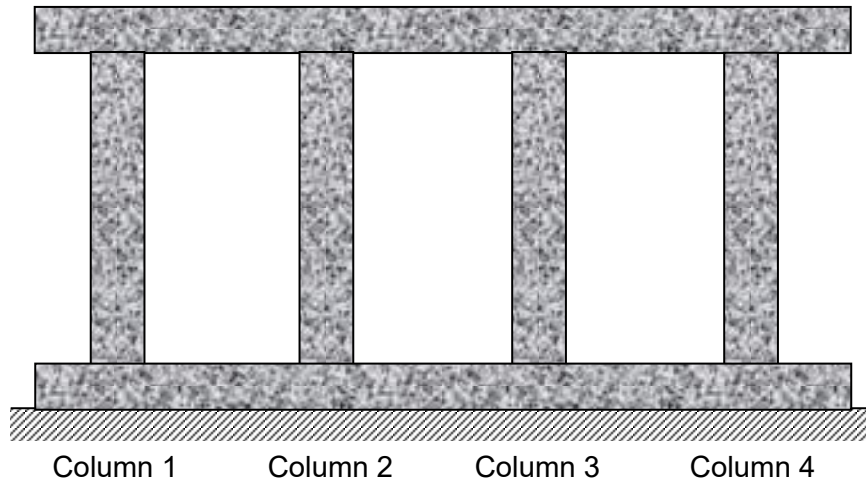
One thing to keep in mind in selecting a controlling defect in a tie is the relative extent of the defects over the element. An example would be a situation where there is a steel girder with cracking and corrosion defects, both in Condition State 3. The cracking occurs in the last foot of the element, the corrosion over the last 10 feet. The two defects occur together only in the last foot. If the inspector decides corrosion controls, then 10 feet of Condition State 3 corrosion is reported, the cracking is not reported at all. If the inspector decides cracking controls, 9 feet of Condition State 3 corrosion and one foot of Condition State 3 cracking are reported. The second solution may be more desirable.

Elements with a unit of "Each" are handled differently. "Each" elements are viewed as a group of indivisible individuals. For example, with columns there can be one column, there can be two columns, there can't be 1.5 columns. Each individual column is entirely in one and only one condition state. Each column can have one and only one defect 'rolled-up' to the element. If two Condition State 3 defects occur on an individual column, only one can be reported, regardless of the relative extent of the two defects. The inspector will need to decide which is the bigger "threat" for that column and report that defect on that column.

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An example may be in order.



Column 1
Pristine Condition

Column 2
Pristine Condition

Column 3
Number of CS 3 spalls
A large amount of CS 2 cracking

Column 4
A minimal amount of CS 2 spalls
CS 2 cracking over all of column

TOTALS REPORTED

CS 1 = 2 (Columns 1 and 2 – Pristine Condition)
CS 2 = 1 (Column 4 CS 2 spalling controls over CS 2 cracking)
CS 3 = 1 (Column 3 – CS 3 because in this column spalling overrides CS 2 cracking)

Note that all defects/conditions stated above must be reported in the inspection report in a manner that makes comparing the defects reported as simple as possible during the next inspection. That may be a chart, table or a narrative.

What follows are the defects noted in hierarchy format to assist the inspector. As stated previously, inspectors must use their own judgment and use this only as a variable guide.

DEFECT HIERARCHY

What follows are the defects noted in hierarchy format to assist the inspector in determining which defect may take precedence over another when the condition states of two different defects for the same element/area are the same. This will be used for VDOT and AASHTO defects. As stated previously, inspectors must use their own judgment and use this only as a variable guide.

In all instances, the 7000 – Damage defect does not “roll-up” into the total quantity for the condition state; therefore, it has not been shown in the hierarchies given below, but would still be utilized to indicate the presence of such; capture the specific damage caused by the impact under the appropriate material defect entry as well.

Reinforced Concrete

Deck

- 1090 - Exposed Rebar
- 1080 - Delamination/Spall/Patched Area
- 1120 - Efflorescence/Rust Staining
- 1130 - Cracking
- 7000 - Damage
- 1190 - Abrasion/Wear

- 2000 - Deck Drainage System Effectiveness (VDOT Defect only to be used for Element 802 Deck Drains)

Superstructure

- 1900 - Distortion
- 1090 - Exposed Rebar
- 1080 - Delamination/Spall/Patched Area
- 1120 - Efflorescence/Rust Staining
- 1130 - Cracking

Substructure

- 6000 - Scour
- 4000 - Settlement
- 1090 - Exposed Rebar
- 1080 - Delamination/Spall/Patched Area
- 1120 - Efflorescence/Rust Staining
- 1130 - Cracking
- 1190 - Abrasion/Wear

- 6501 - Facing Panel Displacement/Separation (VDOT Defect only to be used for Element 830 MSE Wall)
- 6502 - Facing Panel Bulging (VDOT Defect only to be used for Element 830 MSE Wall)
- 6503 - Facing Panel Joint Opening (VDOT Defect only to be used for Element 830 MSE Wall)

(continues)

(continued)

Culvert

- 6000 - Scour
- 4000 - Settlement
- 1900 - Distortion (To be used for reporting distortion from the original line or grade of the element to include misalignment at construction joints)
- 1090 - Exposed Rebar
- 1080 - Delamination/Spall/Patched Area
- 1120 - Efflorescence/Rust Staining
- 1130 - Cracking
- 1190 - Abrasion/Wear

Prestressed Concrete

Deck

- 1100 - Exposed Prestressing
- 1090 - Exposed Rebar
- 1080 - Delamination/Spall/Patched Area
- 1120 - Efflorescence/Rust Staining
- 1110 - Cracking
- 1190 - Abrasion/Wear

Superstructure

- 1900 - Distortion
- 1100 - Exposed Prestressing
- 1090 - Exposed Rebar
- 1080 - Delamination/Spall/Patched Area
- 1120 - Efflorescence/Rust Staining
- 1110 - Cracking

Substructure

- 6000 - Scour
- 4000 - Settlement
- 1100 - Exposed Prestressing
- 1090 - Exposed Rebar
- 1080 - Delamination/Spall/Patched Area
- 1120 - Efflorescence/Rust Staining
- 1110 - Cracking
- 1190 - Abrasion/Wear

(continues)

(continued)

Culvert

- 6000 - Scour
- 4000 - Settlement
- 1900 - Distortion (To be used for reporting distortion from the original line or grade of the element to include joints open beyond the manufacturer's recommendations and for misalignment at joints between sections)
- 1100 - Exposed Prestressing
- 1090 - Exposed Rebar
- 1080 - Delamination/Spall/Patched Area
- 1120 - Efflorescence/Rust Staining
- 1130 - Cracking
- 1190 - Abrasion/Wear

Steel (All Elements)

- 6000 - Scour (Substructure and Culvert Only)
- 4000 - Settlement (Substructure and Culvert Only)
- 1010 - Cracking
- 1020 - Connection
- 1900 - Distortion (To be used for reporting distortion from the original line or grade of the element)
- 1000 - Corrosion
- 1030 - Misalignment
- 1200 - Wear

Timber

Deck

- 1140 - Decay/Section Loss
- 1180 - Abrasion/Wear
- 1160 - Cracking
- 1170 - Split/Delamination
- 1020 - Connection
- 1150 - Check/Shake

Superstructure

- 1140 - Decay/Section Loss
- 1160 - Cracking
- 1170 - Split/Delamination
- 1020 - Connection
- 1150 - Check/Shake
- 1180 - Abrasion/Wear

(continues)

(continued)

Substructure

- 6000 - Scour
- 4000 - Settlement
- 1140 - Decay/Section Loss
- 1180 - Abrasion/Wear
- 1160 - Cracking
- 1170 - Split/Delamination
- 1150 - Check/Shake
- 1020 - Connection

Culvert

- 6000 - Scour
- 4000 - Settlement
- 1140 - Decay/Section Loss
- 1180 - Abrasion/Wear
- 1160 - Cracking
- 1170 - Split/Delamination
- 1150 - Check/Shake
- 1020 - Connection

Masonry (All Elements)

- 6000 - Scour (Substructure and Culvert Only)
- 4000 - Settlement (Substructure and Culvert Only)
- 1640 - Masonry Displacement
- 1080 - Delamination/Spall/Patched Area
- 1620 - Split/Spall
- 1630 - Patched Area
- 1610 - Mortar Breakdown
- 1900 - Distortion
- 1120 - Efflorescence/Rust Staining

Other (All Elements)

- 6000 - Scour (Substructure and Culvert Only)
- 4000 - Settlement (Substructure and Culvert Only)
- 1900 - Distortion
- 1010 - Cracking
- 1020 - Connection
- 1220 - Deterioration
- 1120 - Efflorescence/Rust Staining
- 1080 - Delamination/Spall/Patched Area
- 1130 - Cracking
- 1000 - Corrosion

(continues)

(continued)

Joints

- 2310 - Leakage
- 2320 - Seal Adhesion
- 2370 - Metal Deterioration or Damage
- 2330 - Seal Damage
- 2340 - Seal Cracking
- 2360 - Adjacent Deck or Header
- 2350 - Debris Impact

- 2399 - Joint Effectiveness (VDOT Defect only to be used for Element 845 – Joint Effectiveness)

Bearings

- 2220 - Alignment
- 2210 - Movement
- 2240 - Loss of Bearing Area
- 2230 - Bulging, Splitting or Tearing
- 1020 - Connection
- 1000 - Corrosion

Wearing Surfaces

- 3230 - Effectiveness
- 3210 - Delamination/Spall/Patched Area/Pothole
- 3220 - Crack

Steel Protective Coating

- 3440 - Effectiveness
- 3420 - Peeling/Bubbling/Cracking
- 3410 - Chalking
- 3430 - Oxide Film Degradation Color/Texture Adherence

Concrete Protective Coating

- 3540 - Effectiveness
- 3510 - Wear

Roadway over Fill

- 3210 - Delamination/Spalls/Patched Area/Pothole (VDOT Defect only to be used for Element 833 Roadway over Fill)
- 3220 - Crack (VDOT Defect only to be used for Element 833 Roadway over Fill)
- 3230 - Effectiveness (VDOT Defect only to be used for Element 833 Roadway over Fill)

(continues)

(continued)

Slope Protection

- 6505 - Unprotected Slope Effectiveness (VDOT Defect only to be used for Element 851 Unprotected Slope)
- 6506 - Paved Slope Effectiveness (VDOT Defect only to be used for Element 852 Protected Slope)
- 6507 - Loss Of Riprap Stone (VDOT Defect only to be used for Element 853 Protected Slope - Riprap)

Channel

- 6508 - Effectiveness Of Stream Control/Channel Protective Devices (VDOT Defect only to be used for Element 854 Channel)
- 6500 - Debris (Channel) (VDOT Defect only to be used for Element 854 Channel)
- 1220 - Deterioration
- 4000 - Settlement
- 6000 - Scour

DEFINITIONS AND GUIDANCE FOR DETERMINING ELEMENTS AND QUANTITIES

GENERAL NOTES

Any item placed to provide support until permanent repairs can be made is not to be entered as a Bridge Element. This includes but is not limited to steel or timber bents.

When backwalls or bearing seats consist of a material different from the primary material of the substructure unit, the condition of the backwall or the bearing seat will not be tracked using a Bridge Element or defect.

The Virginia Division Office of FHWA has advised that the unit of measure for Bridge Management Element 515 (Steel Protective Coating) shall be square feet.

FOOTBRIDGES AND PEDESTRIAN BRIDGES

Footbridges and Pedestrian bridges will be placed in InspectX and NBI ratings are required. However, no element data collection for either type of structure is required. Element Data may be collected if desired by the District.

DECK ELEMENTS

Deck Area

Deck area shall be calculated based on the out-to-out width of the slab and the total length of the bridge (back-to-back of backwalls or end-to-end of slab).

Decks and Slabs

Deck Elements and Slab Elements are defined separately. Care should be taken to define these elements correctly. Slab Elements are stand-alone structures such as slab spans. Primary structural members, such as steel beams or prestressed beams, support Deck Elements.

Decks – Multiple

A bridge may be composed of several deck elements i.e. Element 12 – Reinforced Concrete Deck and Element 28 – Steel Deck, Open Grid. For this example, both elements shall be coded.

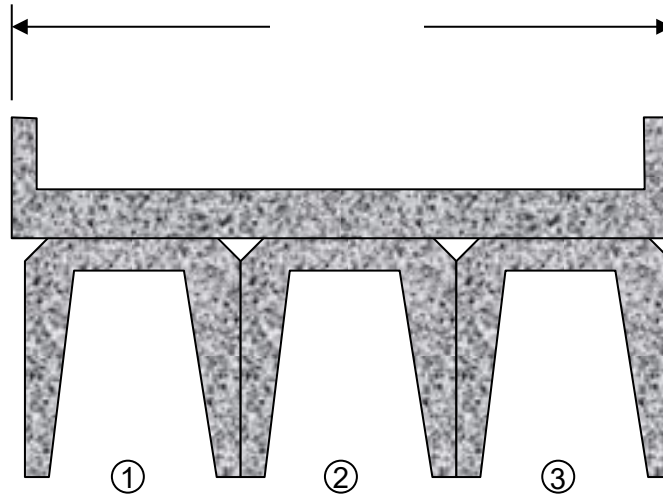
SUPERSTRUCTURE ELEMENTS

Determining Quantities

This convention applies to all girders, channels, and box girders. The quantity is NOT dependent on the number of visible pairs of girder faces but is determined by the length of the bridge multiplied by the number of girders, beams, channels or box girders.

See examples below.

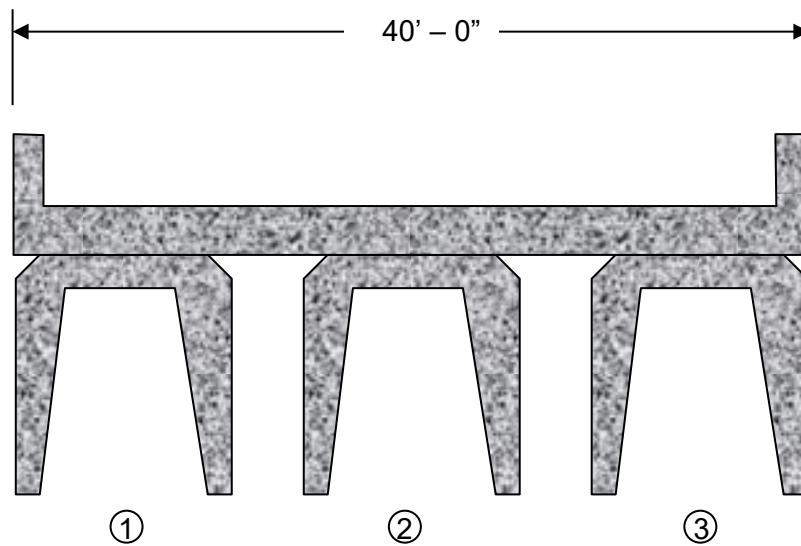
Example 1: Reinforced Concrete Channels (Adjacent)



Since there are three (3) channel beams, the quantity should be the length of the bridge multiplied by 3. For the above figure a bridge that is 25 feet in length and 30 feet wide, the total girder quantity should be 75 feet.

<u>Element #</u>	<u>Element Name</u>	<u>Quantity</u>
12	Reinforced Concrete Deck	750 SF (30 x 25)
110	Reinforced Concrete - Open Girder/Beam	75 LF (3 x 25)

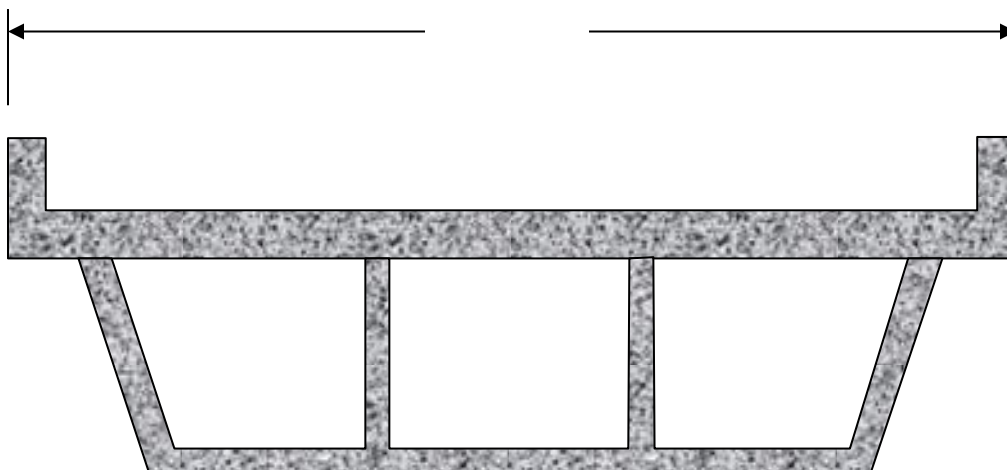
Example 2: Reinforced Concrete Channels (Spread) (No Prestressing)



Since there are three (3) channel beams, the quantity should be the length of the bridge multiplied by 3. For the above figure a bridge that is 25 feet in length and 40 feet wide, the total girder quantity should be 75 feet.

<u>Element #</u>	<u>Element Name</u>	<u>Quantity</u>
12	Concrete Deck	1000 SF (40 x 25)
110	Reinforced Concrete - Open Girder/Beam	75 LF (3 x 25)

Example 3: Cast-in-Place Reinforced Concrete Multi-Cell Box Girders (No Prestressing)



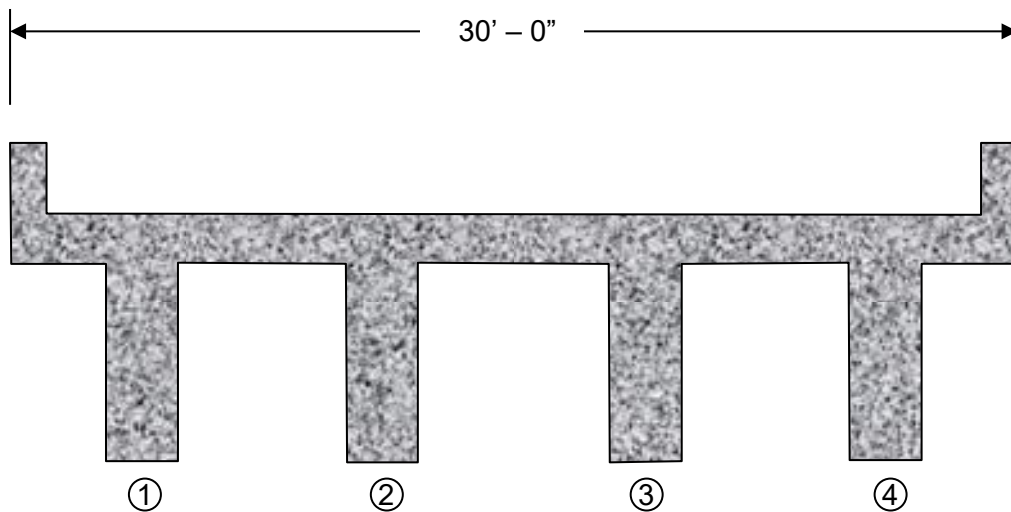
Elements for both the girder and the deck are necessary, even though the unit is integral. A multi-cell box girder is considered to be one girder. For the above figure a bridge that is 25 feet in length and 30 feet wide, the total girder quantity should be 25 feet.

<u>Element #</u>	<u>Element Name</u>	<u>Quantity</u>
16	Reinforced Concrete - Top Flange	750 SF (30 x 25)
105	Reinforced Concrete Closed Web/Box Girder	25 LF (1 x 25)

Note:

If an additional composite structural deck is placed on the beam/girder then a deck element would also apply.

Example 4: Prestressed Concrete Tee Beam



Elements for both the girder and the deck are necessary, even though the unit is integral. The quantity is the length of the bridge multiplied by 4. For the above figure a bridge that is 25 feet in length and 30 feet wide, the total girder quantity should be 100 feet.

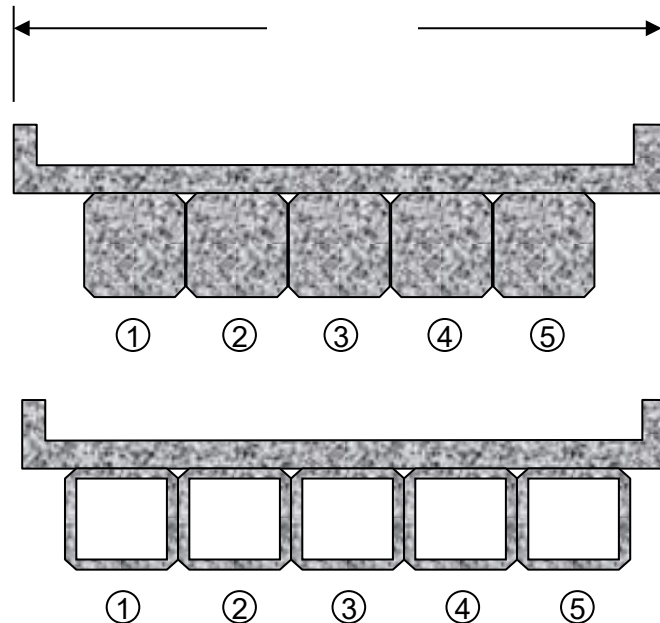
<u>Element #</u>	<u>Element Name</u>	<u>Quantity</u>
15	Prestressed Concrete - Top Flange	750 SF (30 x 25)
109	Prestressed Open Concrete Open Girder/Beam	100 LF (4 x 25)

Note:

If an additional composite structural deck is placed on the beam/girder then a deck element would also apply.

Example 5: Prestressed Concrete Closed Web / Box Girder (Adjacent) (Voided and Un-Voided) with Reinforced Concrete Deck with Mild Steel

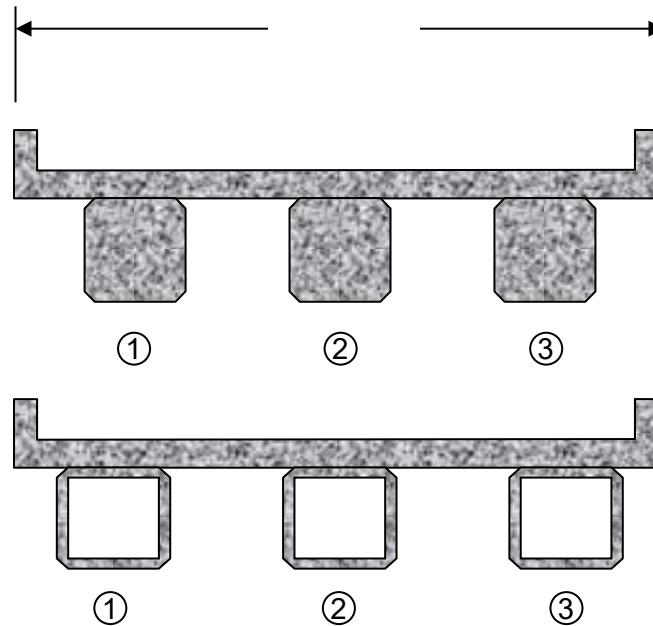
The quantity for box girders is based on the number of girders. A deck element should also be coded.



For each of the above figures a bridge that is 25 feet in length and 30 feet wide, the total girder quantity should be 125 feet.

<u>Element #</u>	<u>Element Name</u>	<u>Quantity</u>
12	Reinforced Concrete Deck	750 SF (30 x 25)
104	Prestressed Concrete Closed Web/Box Girder	125 LF (5 x 25)

**Example 6: Reinforced Concrete Voids and Un-Voids Closed Web / Box Girder (Spread)
(No Prestressing)**

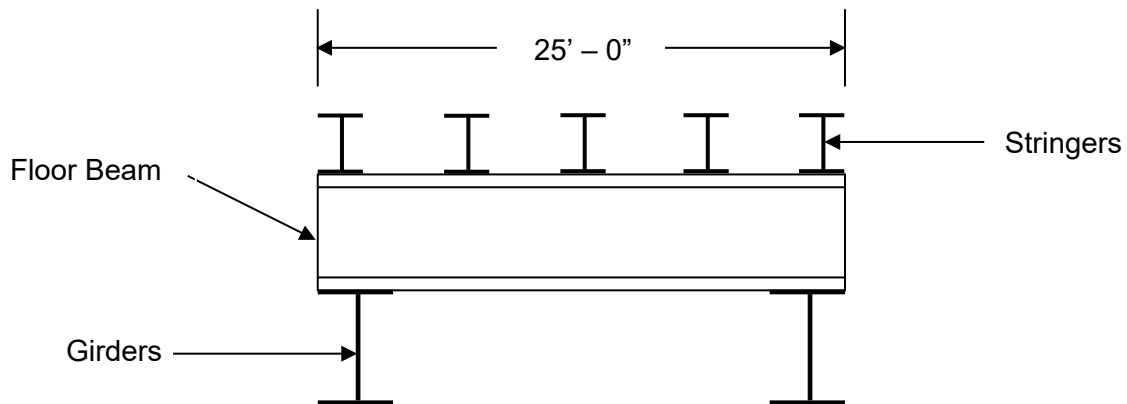


The quantity should be the length of the bridge multiplied by 3. For the above a bridge that is 25 feet in length and 30 feet wide, the total girder quantity should be 75 feet.

<u>Element #</u>	<u>Element Name</u>	<u>Quantity</u>
12	Reinforced Concrete Deck	750 SF (30 x 25)
105	Reinforced Concrete Closed Web/Box Girder	75 LF (3 x 25)

Example 7: Girders / Stringers / Floor Beams

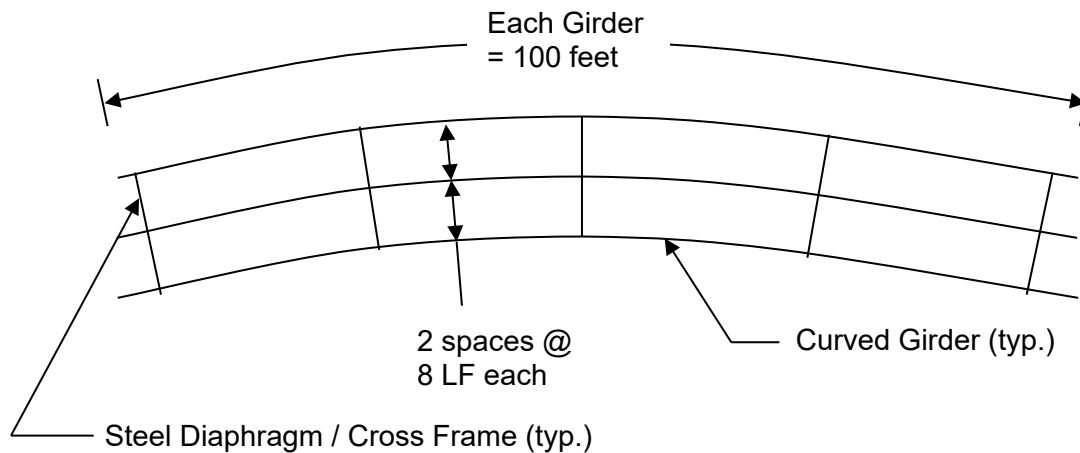
The figure below shows a two-girder bridge that is 50 feet long with three floor beams that are 25 feet long each and five stringers.



<u>Element #</u>	<u>Element Name</u>	<u>Quantity</u>
107	Steel Open Girder/Beam	100 LF (2 x 50)
113	Steel Stringer	250 LF (5 x 50)
152	Steel Floor Beam	75 LF (3 x 25)

Example 8: Steel Diaphragms / Cross Frames for Curved Girder Systems

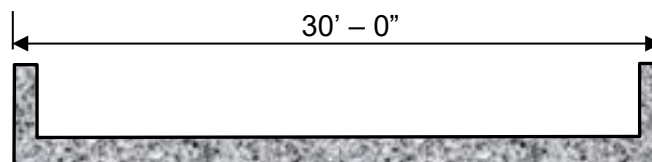
The figure below shows a three-girder curved bridge that is 100 feet long with ten diaphragms that are 8 feet long each.



<u>Element #</u>	<u>Element Name</u>	<u>Quantity</u>
813	Steel Diaphragms / Cross Frames for Curved Girder Systems	80 LF (8 x 10)

The quantity for Element 813 Steel Diaphragms is independent of the length of the girder.

Example 9: Reinforced Concrete Slab



For the above structure the bridge is 25 feet long and 30 feet wide, the element quantities are as follows:

<u>Element #</u>	<u>Element Name</u>	<u>Quantity</u>
38	Reinforced Concrete Slab	750 SF (30 x 25)
331	Reinforced Concrete Bridge Railing	50 LF (2 x 25)

A prestressed concrete slab element could also be used in this example if the slab was of that material type.

SUBSTRUCTURE ELEMENTS

Abutments – General

Abutments will be coded using the AASHTO Elements (including Wings) plus the Virginia Agency Defined Elements for wingwalls (each).

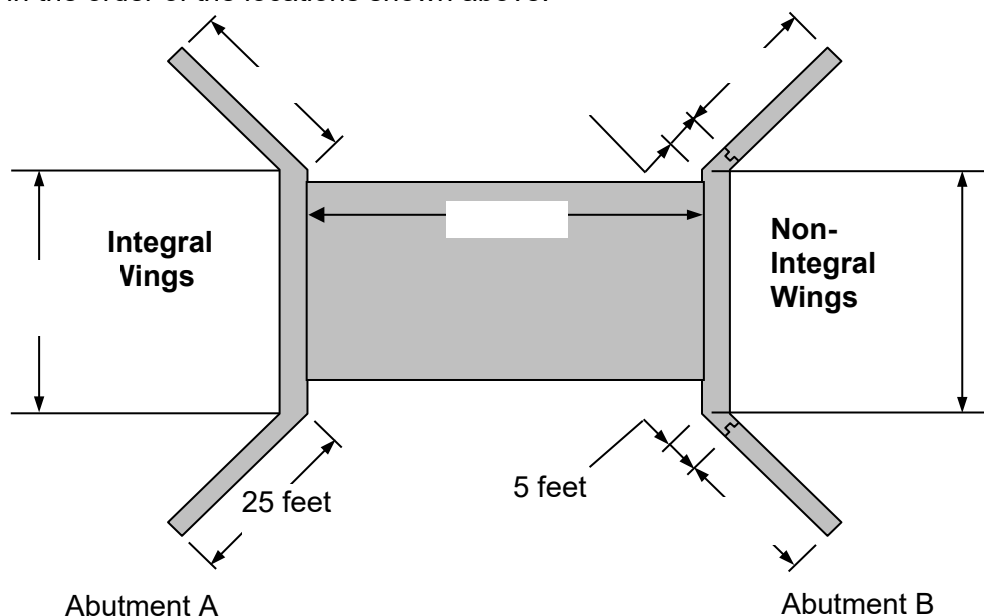
For the purposes of element-level inspections, the condition of backwalls, bearing seats, breastwalls, weep holes, and footings are to be considered as a part of the abutment.

Gravity abutments should be coded as reinforced concrete abutments.

Wings will be considered to begin at the wing tip and end at one of the following locations:

- first vertical construction joint (such as cold joint and water stop) adjacent to the substructure/culvert proper
- break of the wing
- edge of the deck slab

In cases where more than one of the above locations is present on a substructure/culvert element, the wingwalls shall be considered to begin at the wing tip and end at the first location reached in the order of the locations shown above.



Example for Reinforced Concrete Abutments

Abutment A for Integral Wings

<u>Element #</u>	<u>Element Name</u>	<u>Quantity</u>
215	Reinforced Concrete Abutment	75 LF
824	Reinforced Concrete Wingwall	2 Each

Abutment B for Non-Integral Wings

<u>Element #</u>	<u>Element Name</u>	<u>Quantity</u>
215	Reinforced Concrete Abutment	35 LF (25 + 5 + 5)
824	Reinforced Concrete Wingwall	2 Each

Visible/exposed lengths of footings of integral wings shall be included in the Reinforced Concrete Footing element with appropriate defects; quantities of footings of non-integral wings shall not be included.

Defects on the wingwalls shall be included with the Reinforced Concrete Abutment and Wingwall elements when wings are integral per MBEI. For non-integral wings, only defects in the portion from the abutment proper to the vertical joint will be included with the Reinforced Concrete Abutment element and the defects in the remainder of the wing shall be included in the Wingwall element.

Timber Abutments/Bents/Piers/Pile Bents, Etc.

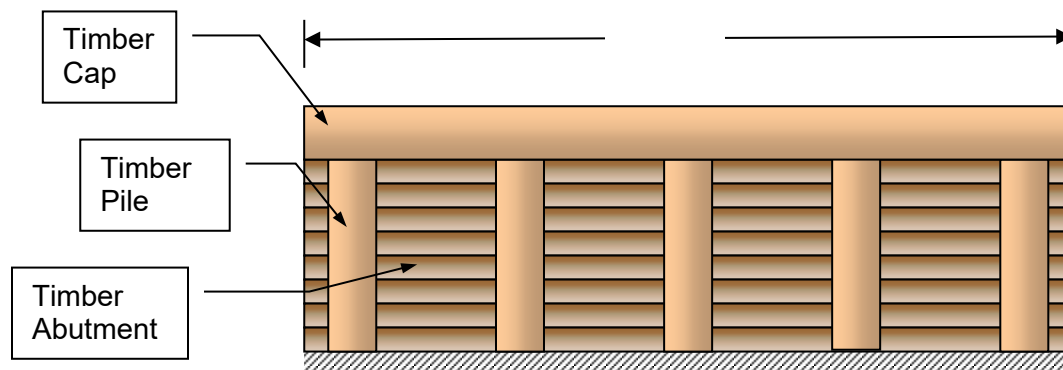
General

The condition of all diagonal/cross bracing on substructure units should be considered in rating the condition of the appropriate substructure unit (columns, piles, etc.). In the event that the bracing is deteriorated, deteriorated sections of bracing should be considered to be part of the nearest appropriate substructure unit (columns, piles, etc.). Considering the Timber Bent shown below, if there was no decay of the piles, but there was decay of the cross bracing with no loss of strength or indication of deflection, then the element 'Timber Pile' should be placed in Condition State 2.

Timber mudsills will be considered with the element 'Timber Pier Cap' and counted separately. In the example on the following page, if a mudsill existed that was 24 feet wide, the quantity for the element 'Timber Pier Cap' would be 48 feet. Note that the element 'Timber Column' shall be used for members between a cap and a mudsill.

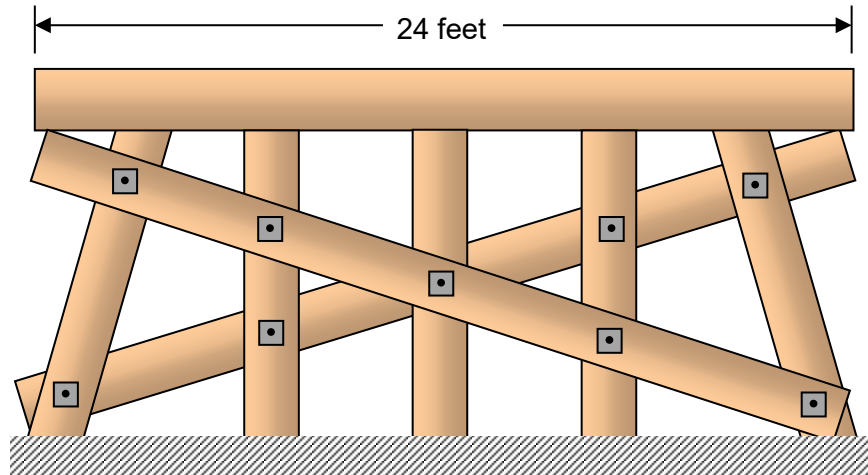
Timber Abutments

Typically coded as 3 different elements: a timber cap, timber piles, and a timber abutment. The timber abutment in these cases will consist only of the timber lagging of the abutment.

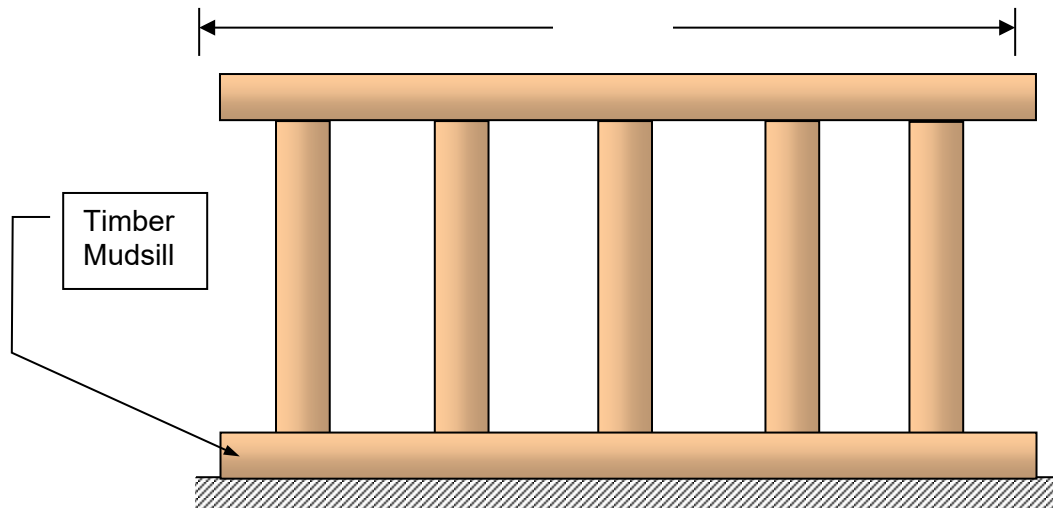


<u>Element #</u>	<u>Element Name</u>	<u>Quantity</u>
216	Timber Abutment	29 LF
228	Timber Pile	5 EA
235	Timber Cap	29 LF

Timber Bents/Piers



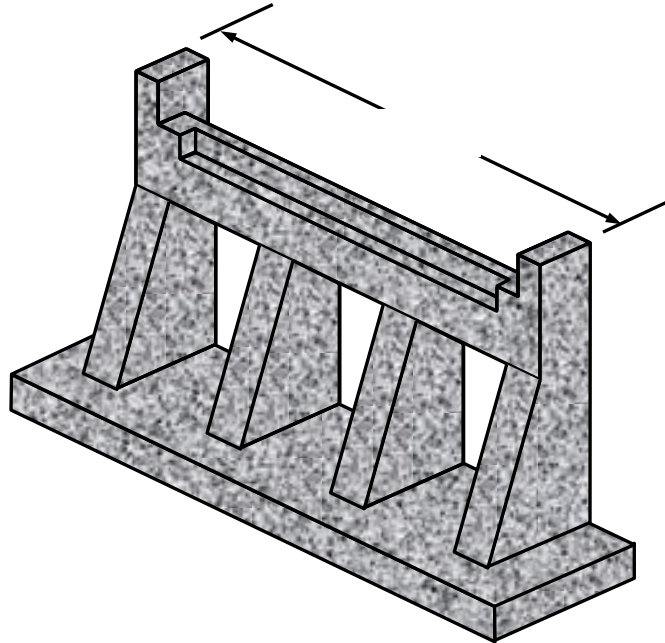
<u>Element #</u>	<u>Element Name</u>	<u>Quantity</u>
228	Timber Pile	5 EA
235	Timber Pier Cap	24 FT



<u>Element #</u>	<u>Element Name</u>	<u>Quantity</u>
206	Timber Column	5 EA
235	Timber Pier Cap (includes Mudsill)	48 FT (2 x 24)

Spill-Through Abutments

Whether the columns are visible or not, the abutment should be coded as indicated below. If the columns are present but not directly visible, reasonable efforts should be made to observe/inspect the columns so that defects may be identified and recorded. Destructive or non-destructive testing is not necessarily needed. If the footing was visible, then the Reinforced Concrete Footing element shall be coded with appropriate defects.

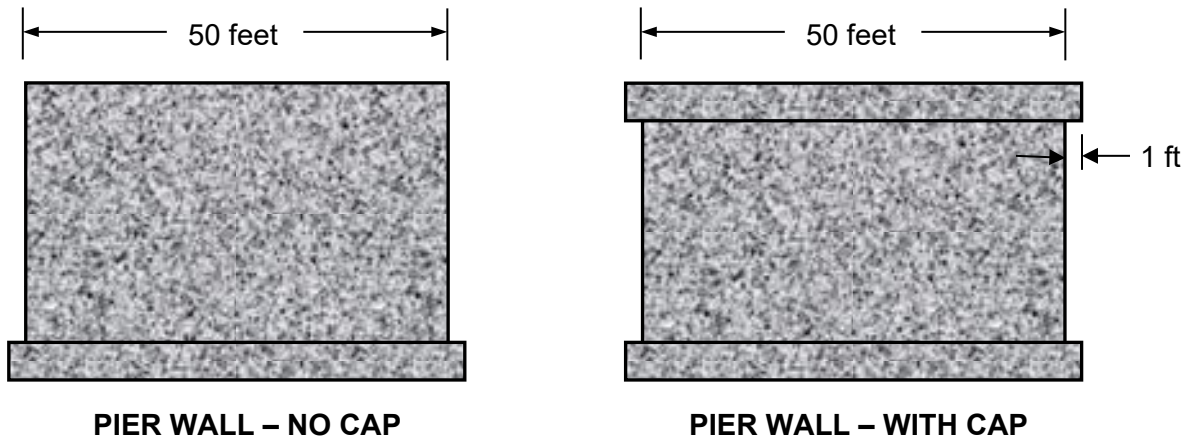


<u>Element #</u>	<u>Element Name</u>	<u>Quantity</u>
205	Reinforced Concrete Column	4 EA
234	Reinforced Concrete Pier Cap	25 LF

Piers – General

The following piers will have two elements consisting of a cap and either columns/piles or pier walls. If the footing was visible, then the Reinforced Concrete Footing element shall also be coded.

Where there is no clear distinction between the column/pier wall and a cap if there is no cap.



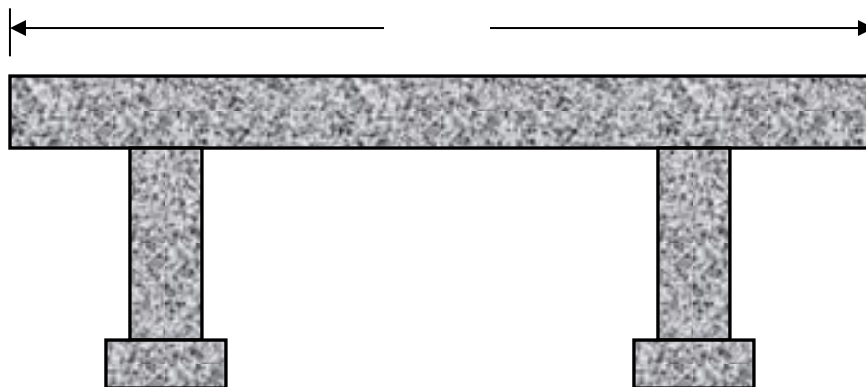
Example on right above – Pier Wall – With Cap

<u>Element #</u>	<u>Element Name</u>	<u>Quantity</u>
210	Reinforced Concrete Pier Wall	50 LF
234	Reinforced Concrete Cap	52 LF

Example on left above – Pier Wall – No Cap

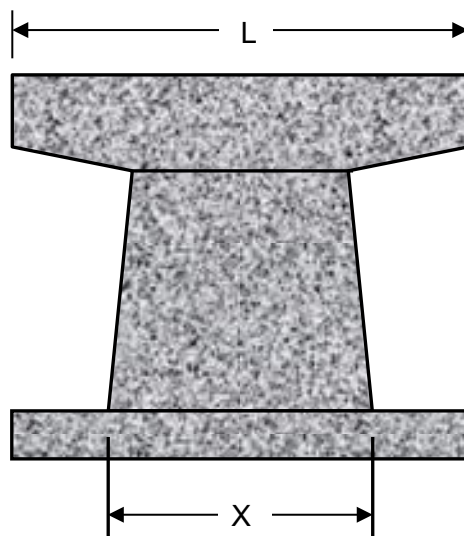
<u>Element #</u>	<u>Element Name</u>	<u>Quantity</u>
210	Reinforced Concrete Pier Wall	50 LF

Column Bent



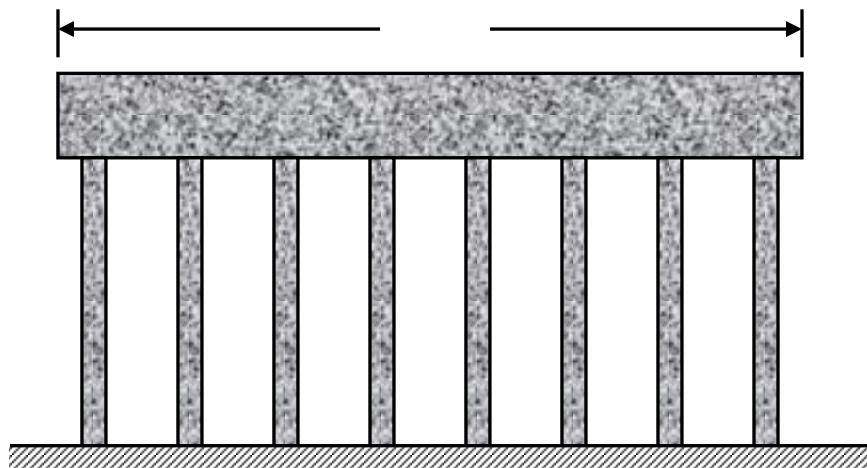
<u>Element #</u>	<u>Element Name</u>	<u>Quantity</u>
205	Reinforced Column	2 EA
234	Reinforced Concrete Cap	28 LF

Hammer Head Pier



<u>Element #</u>	<u>Element Name</u>	<u>Quantity</u>
205	Reinforced Concrete Column (If X is less than 10 feet)	1 EA
210	Reinforced Concrete Pier Wall (If X is equal to or greater than 10 feet)	X FT
234	Reinforced Concrete Pier Cap	L FT

Pile Bent

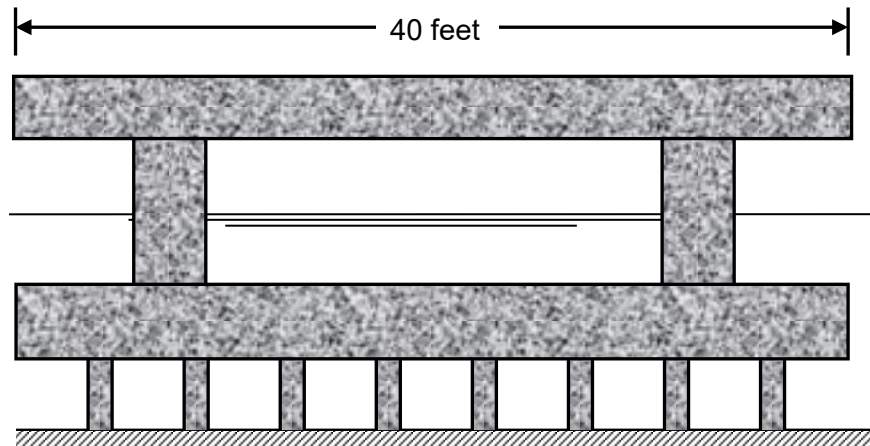


<u>Element #</u>	<u>Element Name</u>	<u>Quantity</u>
227	Reinforced Concrete Pile	8 EA
234	Reinforced Concrete Cap	40 LF

Prestressed concrete elements could also be used in this example.

Column Bent With Low-Water Pile Cap

As shown below there can be cases where reinforced concrete low-water pile caps are used and may or may not be submerged. In these cases, they should be coded using Element 220 Reinforced Concrete Pile Cap/Footing.

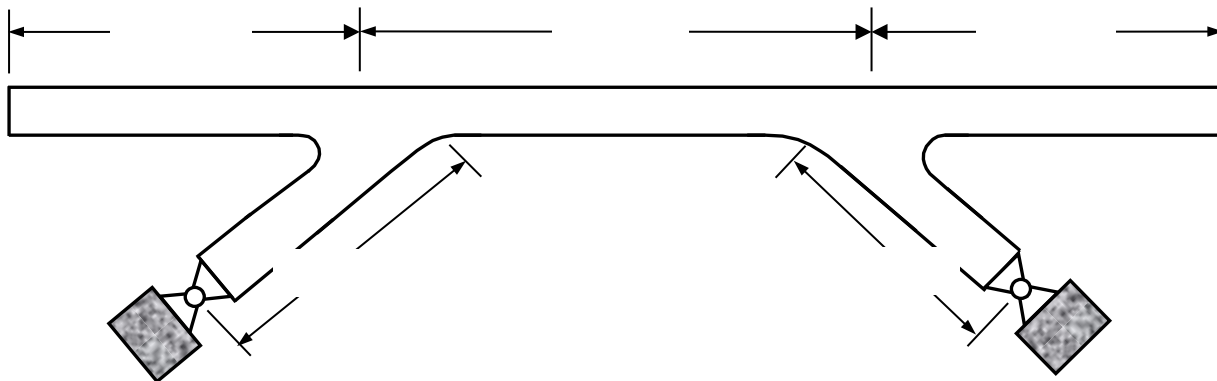


<u>Element #</u>	<u>Element Name</u>	<u>Quantity</u>
205	Reinforced Concrete Column	2 EA
220	Reinforced Concrete Pile Cap/Footing	40 LF
227	Reinforced Concrete Pile	8 EA
234	Reinforced Concrete Cap	40 LF

Prestressed concrete elements could also be used in this example.

RIGID FRAMES AND THREE SIDED STRUCTURES

Steel

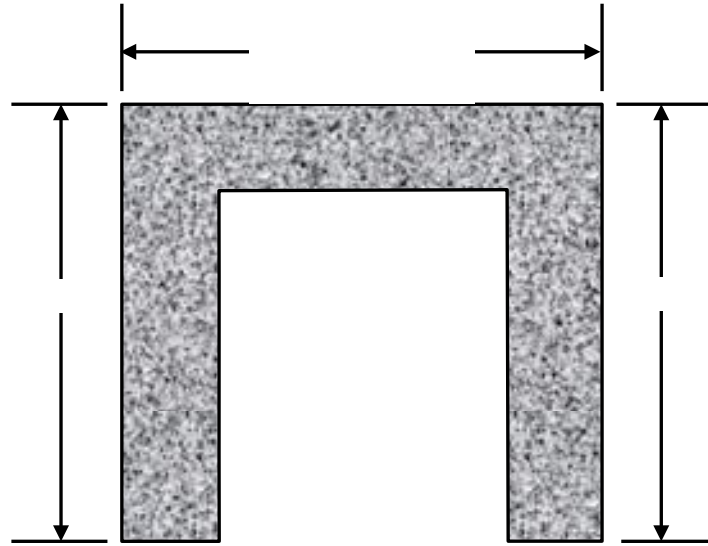


<u>Element #</u>	<u>Element Name</u>	<u>Quantity</u>
107	Steel Open Girder/Beam	450 LF (per girder line)

The legs of the steel frame in this example are included in the overall length of the element 'Steel Open Girder/Beam'. Any elements located above the bearings are part of the superstructure.

Concrete

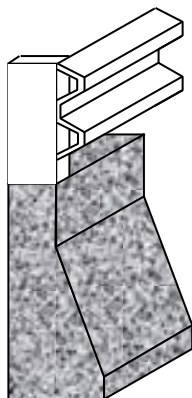
Concrete frames should be coded using the elements for reinforced concrete abutment and reinforced concrete slab.



<u>Element #</u>	<u>Element Name</u>	<u>Quantity</u>
38	Reinforced Concrete Slab	520 SF (20 x 26)
215	Reinforced Concrete Abutment	52 LF (2 x 26)

BRIDGE RAILING, CURBS, SIDEWALKS, MEDIANS, AND TERMINAL WALLS

1. The quantity for bridge railing, curbing, sidewalks, medians and terminal walls are measured from end of structure length to end of structure length.
2. Sidewalks are defined as being 3 feet wide or greater. This dimension distinguishes the sidewalk from the safety curb.
3. Sidewalks and medians 3 feet wide or greater should be coded as a sidewalk, and included in the sidewalk element.
4. Curbs, sidewalks and medians less than 3 feet in width should be coded as concrete railing and measured in linear feet along the centerline of the bridge.
5. Chain link pedestrian fencing used as safety fence on a structure shall be included with the linear feet of any other steel railing in place. If no other steel railing is in place it shall be coded as a separate steel railing. For example, if a 100 foot long structure has steel railing and a chain link pedestrian fence on each side of the structure, 200 linear feet shall be coded for Element 330 Metal Bridge Railing. If no other steel railing is in place and there is a chain link pedestrian fence and a concrete bridge railing on each side of the structure, 200 linear feet shall be coded for Element 330 and 200 linear feet shall be coded for Element 331 Reinforced Concrete Bridge Railing.



Assuming the structure being considered is 70 feet long and there is the railing shown above on each side of the structure, the following shall be coded.

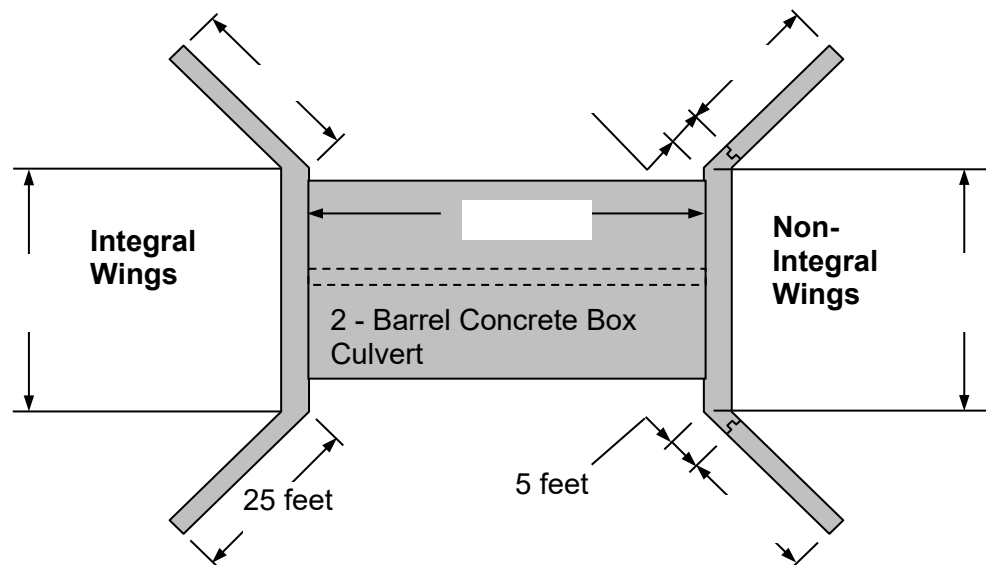
<u>Element #</u>	<u>Element Name</u>	<u>Quantity</u>
330	Metal Bridge Railing	140 LF
331	Reinforced Concrete Bridge Railing	140 LF

CULVERTS

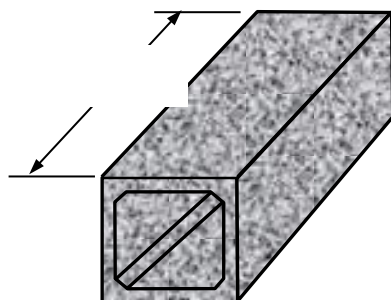
Wingwalls/Endwalls

Element 831 – Culvert/Structure Under Fill Endwall/Headwall are to be used for the recording of culvert endwalls/headwalls.

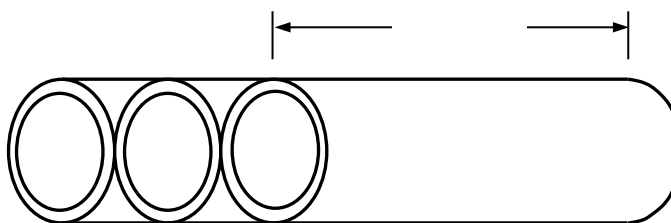
Elements 822, 824, 826, or 828 are to be used for the recording of culvert wingwalls for the appropriate material.



<u>Element #</u>	<u>Element Name</u>	<u>Quantity</u>
241	Reinforced Concrete Culvert	2 x 50 feet = 100 LF
824	Reinforced Concrete Wingwall (Use element appropriate for the wingwall material)	4 Each
831	Culvert/Structure Under Fill Endwall/Headwall	2 Each



Reinforced Concrete Box Culvert



Steel Pipe Culvert

Reinforced Concrete Box Culvert

<u>Element #</u>	<u>Element Name</u>	<u>Quantity</u>
241	Reinforced Concrete Culvert	50 LF
833	Roadway over Fill	1 EA

Multiple Steel Pipe Culverts

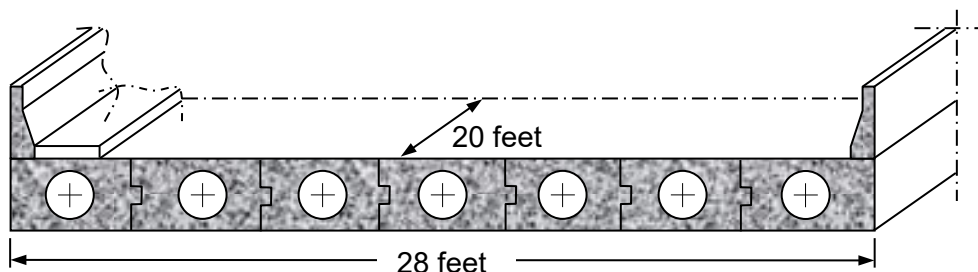
<u>Element #</u>	<u>Element Name</u>	<u>Quantity</u>
240	Steel Culvert	3 x 30 = 90 LF
833	Roadway over Fill	1 EA

Element 833 – Roadway over Fill shall be used to report the condition of any type of roadway over a culvert.

The condition of aluminum culverts shall be reported using Element 243 – Other Culvert.

PRESTRESSED CONCRETE CLOSED WEB/BOX GIRDERS

Element 104 – Prestressed Concrete Closed Web/Box Girders is for both voided and unvoided Prestressed Concrete Girders/Slab Units. A top flange element is also needed. If there is a separate wearing surface it must also be coded so the riding surface can be assessed. Where the girders are not spread and the traffic rides directly on the structural element, regardless of the wearing surface, evaluation of the top flange is considered with Element 15 – Prestressed Concrete Top Flange or Element 16 – Reinforced Concrete Top Flange. This configuration will NOT typically have a separate deck element.

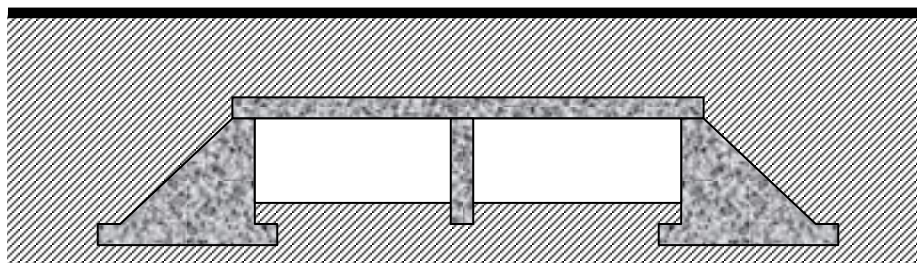


<u>Element #</u>	<u>Element Name</u>	<u>Quantity</u>
15	Prestressed Concrete Top Flange	560 SF (20 x 28)
104	Prestressed Concrete Closed Web/Box Girders	140 LF (7 x 20)
331	Reinforced Concrete Bridge Railing (2 feet wide each)	40 LF (2 x 20)
510	Wearing Surface	420 SF (20 x 21)
801	Sidewalk (3 feet wide)	60 SF (3 x 20)

SLAB SPANS COVERED WITH FILL

Even though these structures are covered with fill, deck, superstructure, and substructure elements should be defined as appropriate. The appropriate slab element based upon material type should be used to report the condition of the slab. Element 833 Roadway over Fill shall be used to report the condition of any roadway over fill.

All or part of the top surface will not be visible for inspection, therefore the element shall be assessed based on the available visible surface, top and/or bottom of the slab. If both top and bottom surfaces are not visible, the condition shall be assessed based on destructive and/or non-destructive testing or indicators in the materials covering the surfaces. Information from other sources may be helpful in coding this element.

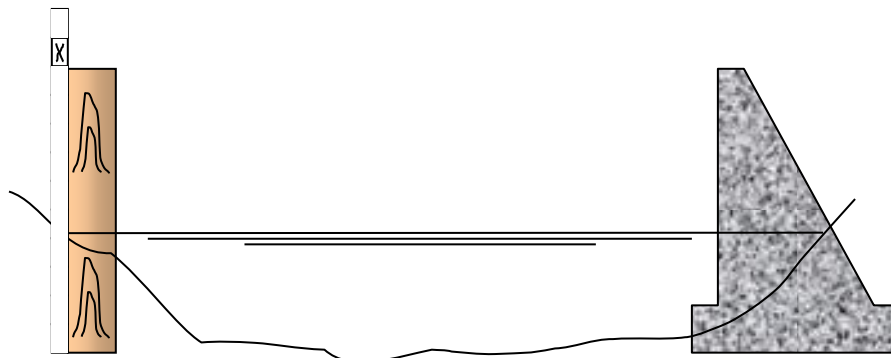


<u>Element #</u>	<u>Element Name</u>	<u>Quantity</u>
38	Reinforced Concrete Slab	Area in SF
210	Reinforced Concrete Pier Wall	Length in LF
215	Reinforced Concrete Abutment	Length in LF
833	Roadway over Fill	1 EA

SLOPE ELEMENTS

The slope at each abutment, as applicable, should be considered as a separate unit. Therefore, the quantity for slope will typically be '2'.

For the example shown below where the stream extends to the substructure there will be no slope element.



TRUSSES

Determining Quantities

The superstructure of a truss bridge usually consists of two parallel trusses. The quantity is determined by calculating the combined length in linear feet of the parallel trusses. Diagonals, verticals, or cross bracing are not counted as additional quantities.

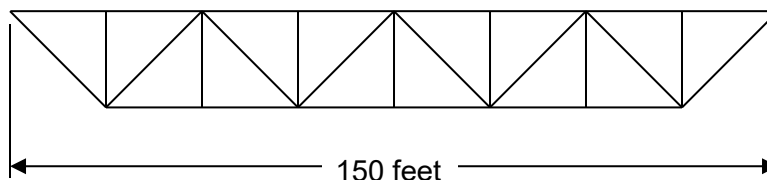
Total Length

The overall quantity is determined by the length of the bridge multiplied by the number of trusses in each span. See example below.

Deck Truss

NOTE

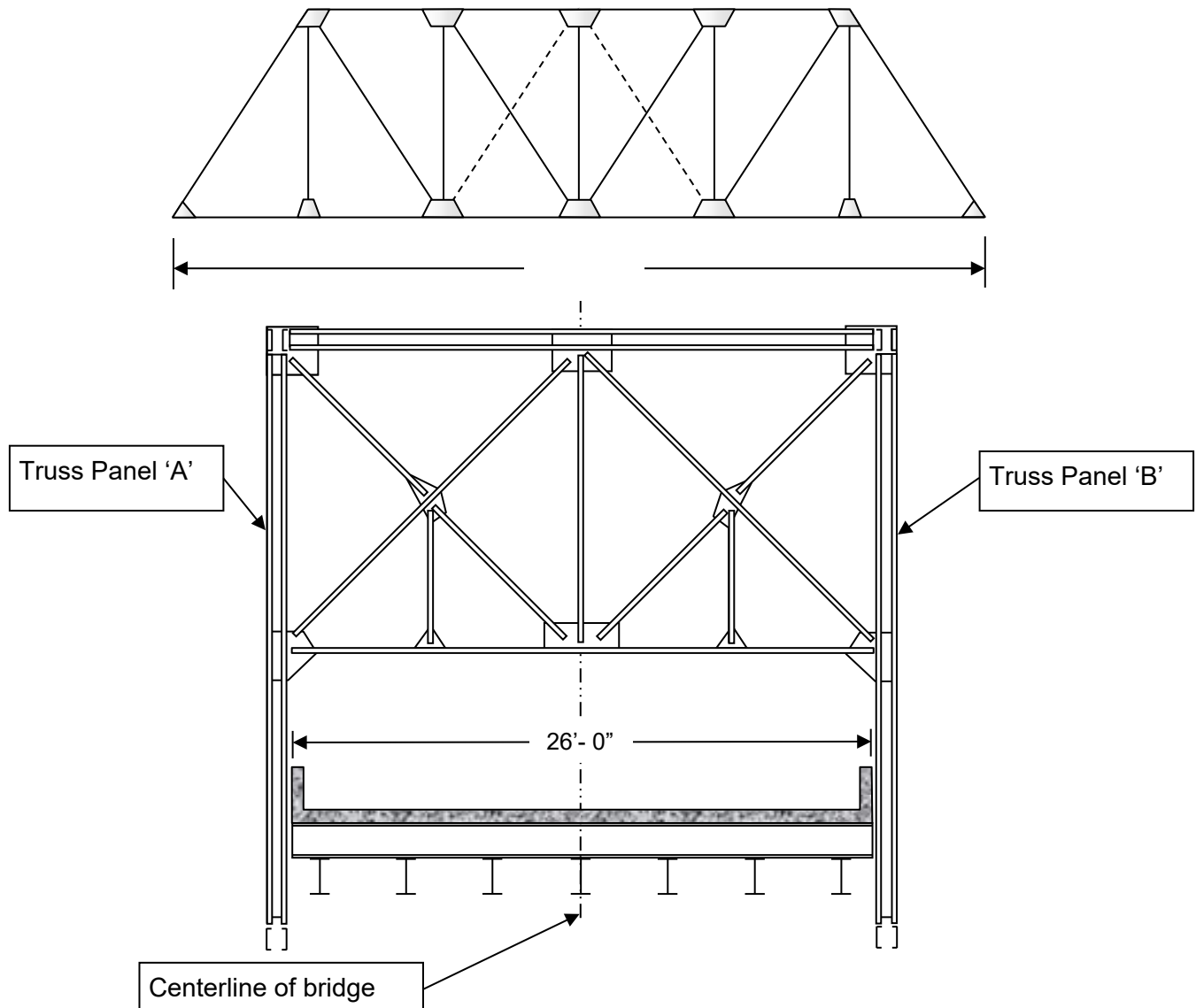
Both through and deck trusses are coded using the same element depending on the material type. The following example is a steel deck truss.



<u>Element #</u>	<u>Element Name</u>	<u>Quantity</u>
120	Steel Truss (Regardless of type and protective system)	300 LF (2 x 150)

Through Truss (includes Pony Trusses)

The following example is for a steel through truss.



<u>Element #</u>	<u>Element Name</u>	<u>Quantity</u>
12	Reinforced Concrete Deck	3900 LF (150 x 26)
113	Steel Stringer	1050 LF (7 x 150)
120	Steel Truss	300 LF (2 x 150)
152	Steel Floor Beam	182 LF (7 x 26)
162	Steel Gusset Plate	24 EA (2 x 12)

The length of the floor beam shown in this example is for illustration purposes only. The actual length of the floor beam should be measured in the field or determined from the plans.

The above illustration and table is not meant to signify all the elements required to be reported.

Deterioration

All deterioration is measured along the length of the bridge.

Diagonals

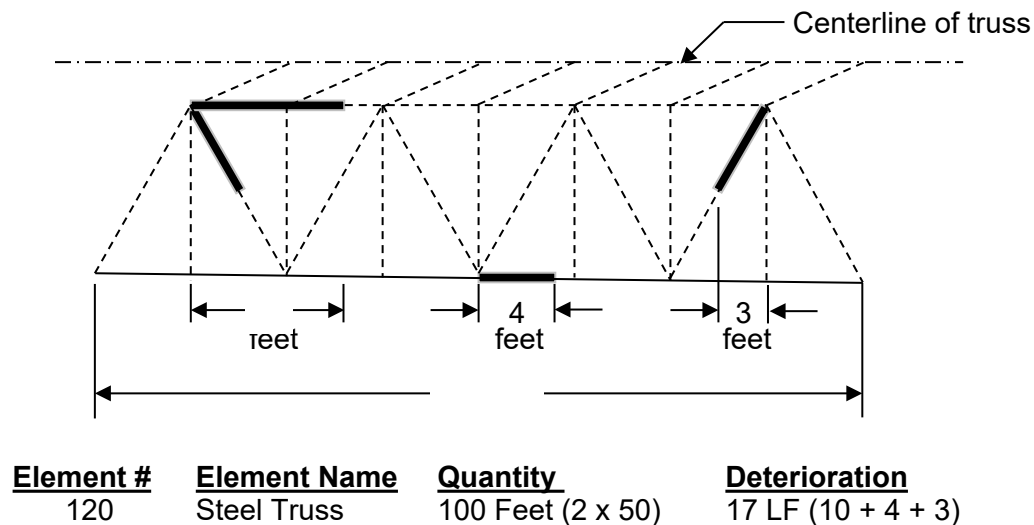
Deterioration of a diagonal member is measured along the length of the bridge and not along the diagonal. The quantity of a deteriorated section in diagonals and verticals that overlap deteriorated section(s) in other parts of the truss should not be double-counted. See example below.

Portals/Bracing

The quantities of deterioration of portals/bracing should be counted with the truss element. The quantity of a deteriorated section in portals/bracing that overlaps a deteriorated section(s) in other parts of the truss, with which it has been associated, should not be double-counted. The darkened areas in the sketch below indicate areas of deterioration on a member.

Through Truss (includes Pony Trusses)

The following example is a steel through truss. Quantities of deterioration should include both the left and right side trusses. Note that the deterioration should be measured along the horizontal projection. The severity of the deterioration is indicated by using the appropriate condition state. The quantities listed are the deteriorated portions of the truss.



WEIGH SCALE PITS

Scale Pits are no longer inspected by VDOT. The Department of Motor Vehicles (DMV) now has responsibility for the Weigh Scale Pits.

ARCHES

The quantity for the arch element is the sum of the lengths of the arches measured longitudinally from springline to springline along the travel way of the facility carried. This includes all CON/SPAN® and BEBO® (Precast Bridges) and BridgeCor® (Plate Bridges) and similar structure types. See examples below.

For Steel Through Arches, Tied Arches, Deck Arches, etc. there are no separate elements for verticals and diagonals. They shall be included in the arch element i.e. they shall be assessed similar to a truss. Also include other associated steel superstructure elements such as: girders, floor beams, stringers, cables, etc.

The condition of aluminum arches shall be reported using Element 142 – Other Arch.

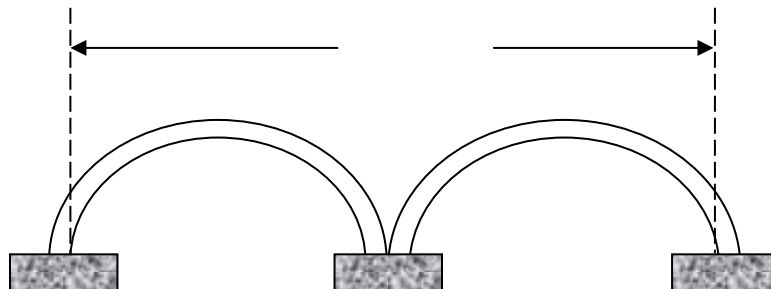
Arches under fill shall be treated in same manner as those not under fill as described above with the following additional considerations:

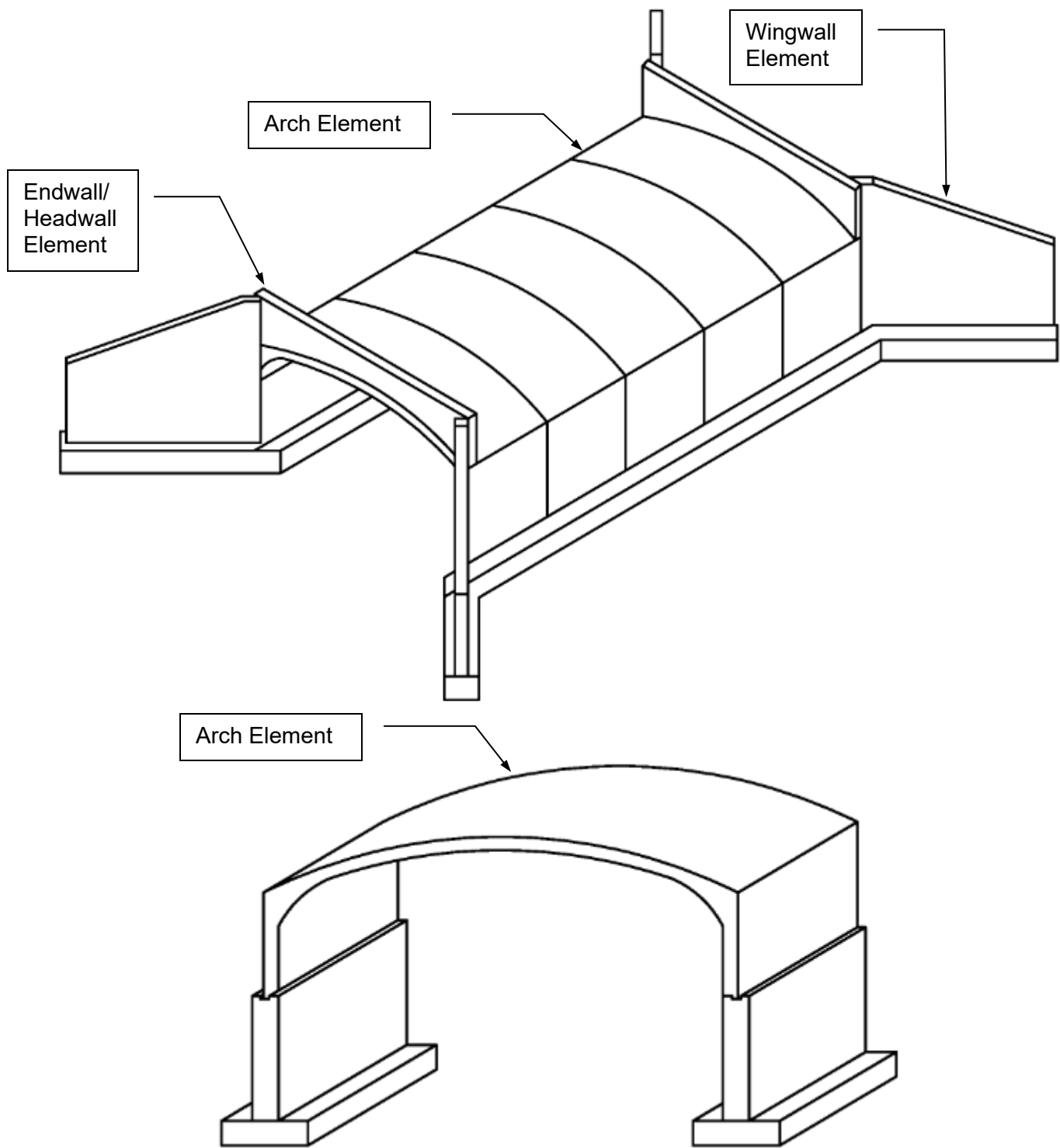
Element 831 – Culvert/Structure Under Fill Endwall/Headwall are to be used for the recording of arch under fill endwalls/headwalls.

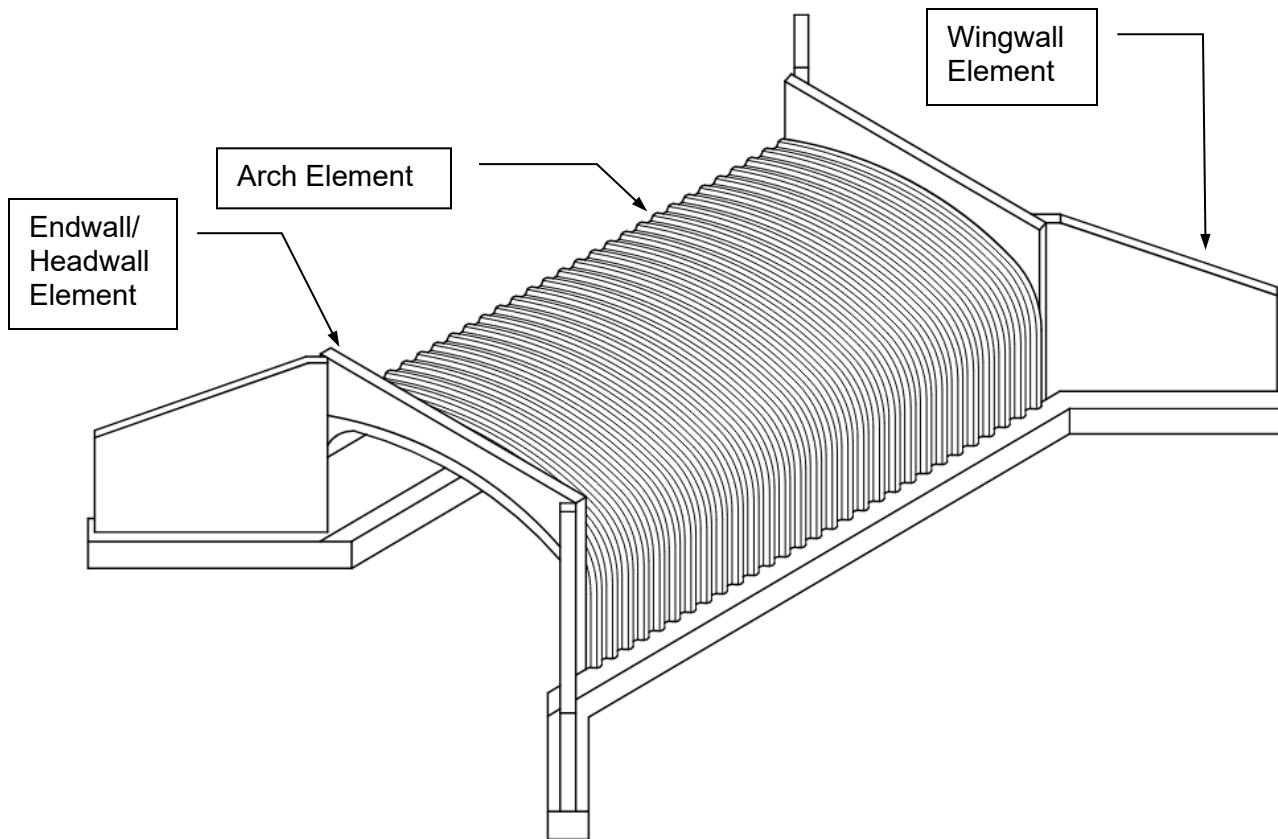
Elements 822, 824, 826, or 828 are to be used for the recording of arch under fill wingwalls for the appropriate material.

Element 833 – Roadway over Fill shall be used for the recording of any type of roadway over an arch under fill.

In Virginia, arches include 'culverts' with no monolithically poured/constructed bottom.







WEARING SURFACE

Wearing surfaces will be coded using the AASHTO element for Wearing Surfaces (Element 510) to record the condition of the wearing surface. Wearing surfaces may also be coded for approach slabs.

For something to be called a wearing surface, it must be placed separately from the time the deck is placed. This is the first and most important test. Therefore, monolithically placed concrete (the $\frac{1}{2}$ " that designers call a wearing surface) is not a wearing surface. The following are examples of wearing surfaces: asphaltic concrete overlays, cementitious overlays, and epoxy overlays. There are others not mentioned here.

EXCEPTIONS - as defined above for something to be called a wearing surface, it must be placed separately from the time the deck is placed. However, if an application is placed separately from the time the deck is placed and if the design engineer or load rating engineer considered it in the capacity (strength) of the structure, it will not be considered an overlay. Therefore, reinforced cementitious overlays, the concrete placed over prestressed deck panels, and other similar applications are not wearing surfaces and are to be considered part of the deck. See previous box girder example.

While this is in conflict with some of the wording of the Bridge Inspector's Reference Manual (BIRM), the above definitions best suit Virginia's need for this information and will reduce or eliminate coding disparities.